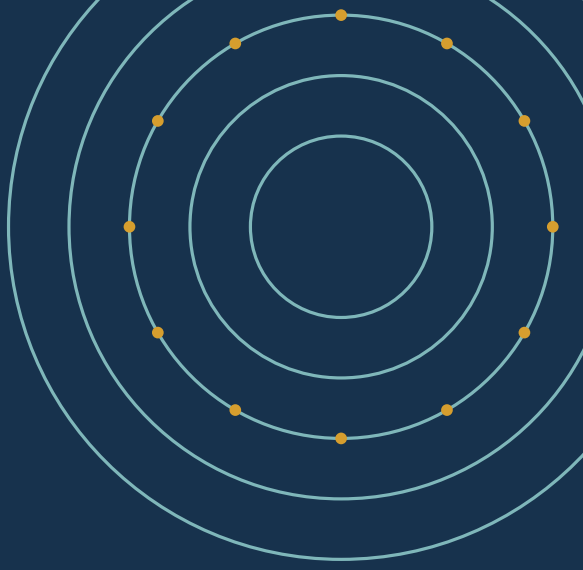
AN INTRODUCTION TO NUMBER THEORY



Patterns, Proofs, and Primes

A visual, inquiry-driven course with exercises

One-semester course packet

Euclid → continued fractions → congruences → groups → prime counting

How to use this course

Big idea

Number theory is an unusually good place to learn what proof feels like. The objects are familiar, experiments are cheap, and elementary questions quickly demand new ways of thinking. This course repeatedly follows the same rhythm:

$$\text{compute} \longrightarrow \text{notice} \longrightarrow \text{conjecture} \longrightarrow \text{prove} \longrightarrow \text{reuse}.$$

Audience and prerequisites

This packet is designed for a first proof-oriented course: strong secondary students, first- or second-year undergraduates, or mathematically mature non-majors. Students should be comfortable with algebra, functions, and mathematical induction. No calculus or abstract algebra is required. A short "proof studio" in the first week can supply any missing experience with direct proof, contradiction, and induction.

Learning goals

By the end, students should be able to:

1. use the Euclidean algorithm both computationally and as a source of proofs;
2. prove unique factorization from Euclid's lemma and use prime exponents to prove $\sqrt{2}$ irrational;
3. generate primes with the sieve of Eratosthenes and justify its stopping rule;
4. encode continued-fraction convergents by unimodular 2×2 matrix products, connect them to extended-Euclidean coefficients, and use them as alternating bounds;
5. explain eventual periodicity in rational base-b expansions and state the Lagrange–Galois picture for quadratic surds;
6. calculate in $\mathbb{Z}/n\mathbb{Z}$, solve linear congruences, and use the Chinese remainder theorem;
7. explain Fermat's little theorem through cyclic symmetry and through finite groups;
8. prove basic identities and divisibility statements for binomial coefficients;
9. understand why central binomial coefficients control the density of primes;
10. write short proofs that clearly separate evidence, claim, and justification.

A 14-week route

The design assumes two 75-minute meetings per week. "Studio" means that at least half the meeting is devoted to student work at boards or in small groups.

Week	Core mathematics	Suggested studio or assessment
1	Divisibility; gcd; Euclidean algorithm	Tiling model; proof diagnostic
2	Bézout identity; linear Diophantine equations	Extended Euclid relay
3	Primes; unique factorization; $\sqrt{2} \notin \mathbb{Q}$; the sieve	Sieve studio; repair a flawed proof; Problem Set 1
4	Finite continued fractions; matrix products; extended Euclid; calendar design	Match remainders with convergent coefficients
5	Alternating convergents; periodicity; Pell equations	Möbius-map studio; Problem Set 2
6	Congruences; units; linear congruences	Clock arithmetic and affine ciphers
7	Chinese remainder theorem	CRT construction studio; short quiz
8	Necklace colorings and Fermat's theorem	Orbit counting with physical beads; Problem Set 3
9	Groups, subgroups, cyclic groups	Cayley-table investigation
10	Lagrange's theorem; group proof of Fermat	Compare two proofs; Problem Set 4
11	Binomial coefficients and Pascal's triangle	Identity stations
12	p-adic valuations; Kummer's carry idea	Color divisibility patterns; Problem Set 5
13	Central binomial coefficient; Chebyshev functions	Prime-counting proof workshop
14	Chebyshev estimate; synthesis	Capstone presentations or oral exam

Assessment design

One workable balance is 45% problem sets, 20% two short proof interviews, 15% computational or visual labs, and 20% capstone. Problem sets should mix three modes:

- **Fluency**: focused calculations that make later proofs readable.
- **Reasoning**: multi-step proofs, often with a hint or intermediate claim.
- **Exploration**: generate data, state a conjecture, then test its limits.

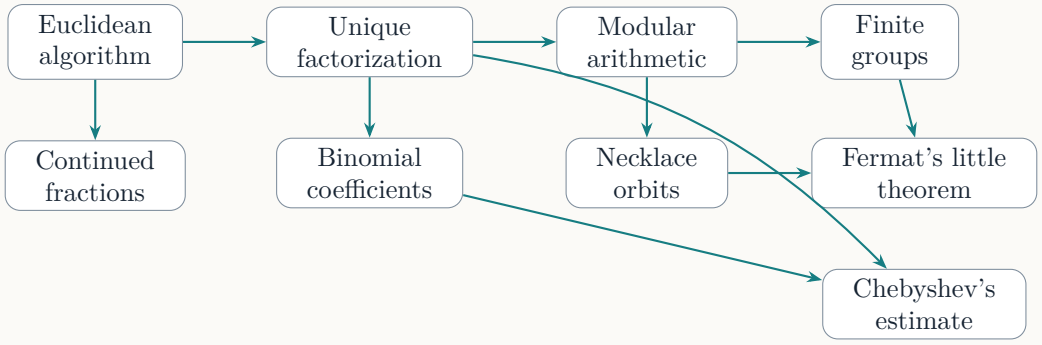
The exercises in each chapter carry these labels. A typical weekly assignment is two fluency problems, two reasoning problems, and one exploration. The appendices supply a short matrix toolkit, continued-fraction arithmetic, an astronomical Venus case study, selected hints and solutions, and reference sheets. An instructor can remove the solution pages when distributing assignments.

Teaching principles

- **Let the picture precede the notation.** Tiling, winding around a clock, rotating a necklace, and counting lattice paths give students something to reason about before symbols compress the idea.
- **Prove one theorem twice.** Fermat's little theorem is the deliberate centerpiece: orbit counting supplies a concrete proof, while groups reveal the reusable structure.

- **Make proof choices visible.** Ask not only "Is this correct?" but "Why did this definition unlock that step?"
- **Use computation as a microscope, not an oracle.** The optional Sage notebook helps students find examples and conjectures; written work must still explain why a pattern continues.

Dependency map



Checkpoint

A good opening prompt. Write 391 and 299 on the board. Ask: "What is the largest square tile that covers a 391×299 rectangle without cutting? How could you be certain?" Do not define gcd until the class has invented the subtraction or division process.

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Chapter 1

The algorithm hidden in a rectangle

Divisibility, gcd, and Bézout's identity

Guiding question 1.1. What is the largest square tile that exactly covers a 391×299 rectangle?

If a square has integer side length d, then it fits precisely only when $d \mid 391$ and $d \mid 299$. We are looking for the *greatest common divisor*, written $\gcd(391, 299)$.

1.1 Divisibility as structure

For integers a, b, we say $a \mid b$ when $b = ak$ for some $k \in \mathbb{Z}$. Divisibility is not an operation like division; it is a yes-or-no relation.

Proposition 1.2 (Division algorithm)**.** *If $a \in \mathbb{Z}$ and $b > 0$, there are unique integers q, r such that*

$$a = bq + r, \qquad 0 \le r < b.$$

The quotient q says how many full copies of b fit inside a; the remainder r is what is left. The crucial observation is

$$\gcd(a, b) = \gcd(b, r) \quad \text{whenever } a = bq + r.$$

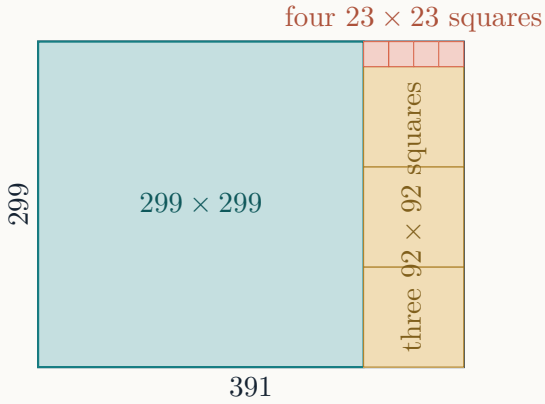
Indeed, an integer divides both a and b exactly when it divides both b and $a - bq = r$.

1.2 Euclid's algorithm

Repeat the division algorithm until the remainder is zero:

$$\begin{aligned} 391 &= 1 \cdot 299 + 92, \\ 299 &= 3 \cdot 92 + 23, \\ 92 &= 4 \cdot 23 + 0. \end{aligned}$$

The last nonzero remainder is 23, so $\gcd(391, 299) = 23$.



The same remainders appear when successively removing the largest possible squares.

Theorem 1.3 (Euclidean algorithm)**.** *For positive integers $a \geq b$, repeated division terminates, and the last nonzero remainder is* $\gcd(a, b)$.

Proof. Each remainder is a nonnegative integer smaller than the previous positive remainder, so the process must stop. Replacing (a, b) by (b, r) does not change the common divisors. At the last line $r_{k-1} = qr_k$, hence the common divisors are precisely the divisors of r_k. □

1.3 Running the movie backward

The remainders do more than find the gcd. Reverse the equations:

$$\begin{aligned} 23 &= 299 - 3 \cdot 92 \\ &= 299 - 3(391 - 299) \\ &= 4 \cdot 299 - 3 \cdot 391. \end{aligned}$$

Thus 23 is an integer linear combination of 391 and 299.

Theorem 1.4 (Bézout's identity)**.** *For integers a, b, not both zero, there are integers x, y such that*

$$ax + by = \gcd(a, b).$$

Moreover, $\gcd(a, b)$ *is the smallest positive integer of the form $ax + by$.*

Corollary 1.5. *The equation $ax + by = c$ has integer solutions exactly when* $\gcd(a, b) \mid c$.

Proof. If $ax + by = c$, every common divisor of a, b divides c. Conversely, multiply a Bézout identity for the gcd by $c/\gcd(a, b)$. □

Common trap

The gcd is positive, but Bézout coefficients need not be. Restricting x, y to positive integers makes the statement false.

Explore

Choose two 3-digit integers. Make a two-column record: Euclid's divisions on the left, backward substitutions on the right. How large can the coefficients become? The optional Sage notebook automates the arithmetic but also asks you to certify its output by substitution.

Exercises

1. Compute $\gcd(2024, 748)$ by Euclid's algorithm. [Fluency]

2. Express $\gcd(527, 341)$ in the form $527x + 341y$. [Fluency]

3. Prove: if $d \mid a$ and $d \mid b$, then $d \mid (ma + nb)$ for all $m, n \in \mathbb{Z}$. [Reasoning]

4. Find all integer solutions to $35x + 22y = 1$. Describe how one solution generates all the others. [Reasoning]

5. Show that $\gcd(a, b) = \gcd(a + kb, b)$ for every $k \in \mathbb{Z}$. Explain why this is the engine of Euclid's algorithm. [Reasoning]

6. A rectangle has integer sides $a > b$. Repeatedly remove the largest square available. Prove that the final square has side $\gcd(a, b)$. [Exploration]

7. Consecutive Fibonacci numbers make Euclid's algorithm take many steps. Test this claim, then prove $\gcd(F_{n+1}, F_n) = 1$. [Exploration]

Chapter 2

Every integer has an address

Primes and the fundamental theorem of arithmetic

Positive integers factor. But why should the factorization into primes be unique? The arithmetic step that makes uniqueness possible is modest enough to miss.

Definition 2.1. An integer $p > 1$ is *prime* if its only positive divisors are 1 and p. An integer $n > 1$ that is not prime is *composite*.

2.1 Euclid's lemma

Lemma 2.2 (Euclid)**.** *If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. Suppose $p \nmid a$. Since the only positive divisors of p are $1, p$, we have $\gcd(p, a) = 1$. Bézout gives $px + ay = 1$. Multiply by b:

$$pbx + aby = b.$$

Both terms on the left are divisible by p, so $p \mid b$. □

Common trap

The word "prime" is essential. We have $6 \mid 2 \cdot 3$, but 6 divides neither factor.

Applying the lemma repeatedly shows: if a prime divides $a_1 a_2 \cdots a_k$, it divides at least one factor.

2.2 Existence and uniqueness

Theorem 2.3 (Fundamental theorem of arithmetic)**.** *Every integer $n > 1$ can be written as a product of primes. Apart from rearranging the factors, this product is unique.*

Existence. Use strong induction on n. If n is prime, we are done. If it is composite, write $n = ab$ with $1 < a, b < n$. By induction, both a and b are products of primes, so n is too. □

Uniqueness. Suppose

$$p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$$

with all p_i, q_j prime. Since p_1 divides the right side, Euclid's lemma says $p_1 = q_j$ for some j. Cancel this common factor. Repeat. No factor can remain on only one side, so the lists agree up to order. □

2.3 An exponent-parity proof of irrationality

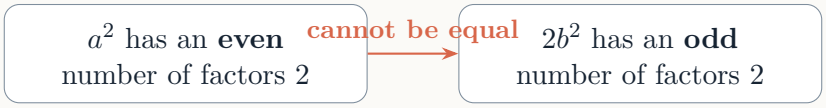
Unique factorization can rule out rational numbers without trying to compute their decimal expansions.

Theorem 2.4. *The number $\sqrt{2}$ is irrational.*

Proof. Suppose instead that $\sqrt{2} = a/b$ for integers a, b with $b \neq 0$. Squaring and clearing the denominator gives

$$a^2 = 2b^2.$$

By the fundamental theorem of arithmetic, the exponent of 2 in the prime factorization of a^2 is even: every exponent occurring in a has been doubled. The exponent of 2 in $2b^2$ is odd: it is one plus twice the exponent occurring in b. Uniqueness says that equal positive integers must have the same exponent of every prime, so this is impossible. □



The contradiction is not merely that one side is even; it is that the two sides contain incompatible numbers of factors 2.

The same proof works with any prime in place of 2. More generally, a rational number can be a perfect square only when every prime exponent in its reduced numerator and denominator is even.

Aside: existence without a witness

Here is a famous one-line use of the law of excluded middle. Set

$$x = (\sqrt{2})^{\sqrt{2}}.$$

Either x is rational or it is irrational. In the first case, the two irrational numbers $a = b = \sqrt{2}$ satisfy $a^b = x \in \mathbb{Q}$. In the second case, the two irrational numbers $a = x$ and $b = \sqrt{2}$ satisfy

$$a^b = \left((\sqrt{2})^{\sqrt{2}}\right)^{\sqrt{2}} = 2.$$

Thus, classically, there exist irrational a and b for which a^b is rational.
The proof is valid in classical mathematics, but it is not a constructive proof: it uses the disjunction "x is rational or x is not rational" without deciding which alternative holds, so it does not identify which of the two proposed pairs is a witness. Constructive and intuitionistic mathematics do not in general accept this unrestricted use of excluded middle.
Unique factorization gives an elementary explicit witness. Take

$$a = \sqrt{2}, \qquad b = \log_{\sqrt{2}} 3.$$

Then $a^b = 3$. Moreover b is irrational: if $b = p/q$ with $q > 0$, then raising $(\sqrt{2})^{p/q} = 3$ to the power $2q$ would give

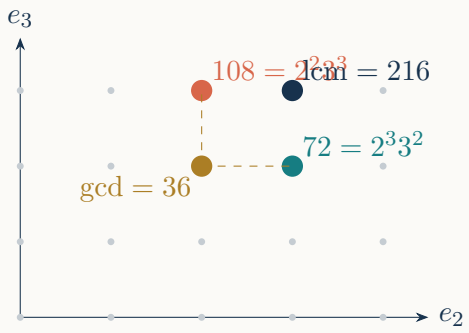
$$2^p = 3^{2q},$$

contradicting unique prime factorization. This time the proof supplies the pair that works. Much more deeply, the Gelfond–Schneider theorem implies that $(\sqrt{2})^{\sqrt{2}}$ is transcendental, so it retrospectively selects the second branch of the classical argument.

Collecting repeated primes gives a coordinate system

$$n = 2^{e_2} 3^{e_3} 5^{e_5} \cdots,$$

where only finitely many exponents are nonzero. Multiplication adds exponent vectors; gcd takes coordinatewise minima; lcm takes coordinatewise maxima.



Prime exponents turn gcd and lcm into geometry.

Proposition 2.5. *If $n = \prod p^{e_p}$ and $m = \prod p^{f_p}$, then*

$$\gcd(n, m) = \prod p^{\min(e_p, f_p)}, \qquad \operatorname{lcm}(n, m) = \prod p^{\max(e_p, f_p)}.$$

In particular, $\gcd(n, m) \operatorname{lcm}(n, m) = nm$.

2.4 The sieve of Eratosthenes

Unique factorization also explains a practical way to find every prime up to a chosen bound N. Write down the integers $2, 3, \ldots, N$. Circle the first uncrossed number, then cross out its larger multiples. Repeat. The circled numbers are precisely the primes.

1	**2**	**3**	~~4~~	**5**	~~6~~	**7**	~~8~~	~~9~~	~~10~~
11	~~12~~	**13**	~~14~~	~~15~~	~~16~~	**17**	~~18~~	**19**	~~20~~
~~21~~	~~22~~	**23**	~~24~~	~~25~~	~~26~~	~~27~~	~~28~~	**29**	~~30~~
31	~~32~~	~~33~~	~~34~~	~~35~~	~~36~~	**37**	~~38~~	~~39~~	~~40~~
41	~~42~~	**43**	~~44~~	~~45~~	~~46~~	**47**	~~48~~	~~49~~	~~50~~

Sieve to 50: the boxed survivors are prime; every composite has been crossed out.

Proposition 2.6 (Why the sieve stops early)**.** *If $n > 1$ is composite, then it has a prime divisor at most $\sqrt{n}$. Consequently, to find all primes up to N, it is enough to cross out multiples of primes $p \le \sqrt{N}$.*

Proof. Write $n = ab$ with $1 < a \le b$. Then $a \le \sqrt{n}$. A prime factor p of a therefore satisfies $p \le a \le \sqrt{n}$. Thus if an integer $n \le N$ survives all sieving primes through $\sqrt{N}$, it cannot be composite. □

Checkpoint

For the display above, only the multiples of $2, 3, 5$, and 7 need to be crossed out. The next prime is $11 > \sqrt{50}$. Notice also that one may begin crossing out at p^2: every smaller multiple of p has already been handled by an earlier prime.

2.5 There is no last prime

Theorem 2.7 (Euclid)**.** *There are infinitely many primes.*

Proof. If $p_1, \dots, p_k$ were all primes, consider $N = p_1 \cdots p_k + 1$. Some prime divides N, but none of the listed primes does: division by any p_i leaves remainder 1. Contradiction. □

This proof does not say that N is prime. It says that N has a prime factor missing from the proposed complete list.

Exercises

1. Factor 7560 and use the exponent vectors to find the number of its positive divisors. [Fluency]
2. Compute $\gcd(2^7 3^2 11, 2^4 3^5 5)$ and the lcm without multiplying first. [Fluency]
3. Prove that if $\gcd(a, b) = 1$ and $a \mid c$, $b \mid c$, then $ab \mid c$. [Reasoning]
4. Reproduce the exponent-parity proof that $\sqrt{2}$ is irrational, then prove that $\sqrt{p}$ is irrational for every prime p. State exactly where unique factorization is used. [Reasoning]
5. Prove that every composite $n > 1$ has a prime divisor at most $\sqrt{n}$. Turn this into a primality test. [Reasoning]
6. Use the sieve of Eratosthenes to list every prime at most 100. Explain why, after sieving by $2, 3, 5$, and 7, no further crossing-out step is needed. [Fluency]
7. Is $n! + 1$ always prime? Find the first counterexample, then explain why Euclid's proof is unaffected. [Exploration]
8. Prove that infinitely many primes are congruent to 3 modulo 4 by adapting Euclid's idea to $4p_1 \cdots p_k - 1$. You may use the fact that a product of numbers congruent to 1 modulo 4 is again 1 modulo 4. [Challenge]
9. Verify directly that $a = \sqrt{2}$ and $b = \log_{\sqrt{2}} 3$ are irrational and satisfy $a^b = 3$. What does this explicit witness add to the excluded-middle proof? [Reasoning]

Chapter 3

Fractions inside fractions

Continued fractions and best approximations

The quotients from Euclid's algorithm are not bookkeeping debris. Read them forward and they produce a new numerical language.

3.1 Euclid, folded up

From

$$415 = 4 \cdot 93 + 43, \quad 93 = 2 \cdot 43 + 7, \quad 43 = 6 \cdot 7 + 1, \quad 7 = 7 \cdot 1,$$

we obtain

$$\frac{415}{93} = 4 + \cfrac{1}{2 + \cfrac{1}{6 + \cfrac{1}{7}}} = [4; 2, 6, 7].$$

Definition 3.1. A finite simple continued fraction is

$$[a_0; a_1, \ldots, a_n] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}},$$

where $a_0 \in \mathbb{Z}$ and $a_i \in \mathbb{N}$ for $i \geq 1$.

The truncations $[a_0; \ldots, a_k] = p_k/q_k$ are called *convergents*. They can be computed without rebuilding the nested fraction:

$$p_k = a_k p_{k-1} + p_{k-2}, \qquad q_k = a_k q_{k-1} + q_{k-2},$$

with $p_{-2} = 0, p_{-1} = 1, q_{-2} = 1, q_{-1} = 0$.

3.2 The matrix engine

Package one continued-fraction digit into the matrix

$$D(a) = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}.$$

Repeated matrix multiplication stores every convergent at once:

$$D(a_0)D(a_1)\cdots D(a_k) = \begin{pmatrix} p_k & p_{k-1} \\ q_k & q_{k-1} \end{pmatrix}. \tag{3.1}$$

Indeed, multiplying the right-hand side by $D(a_{k+1})$ produces exactly the two convergent recurrences. The first column advances to the next convergent, while the old first column becomes the second.

Each digit matrix has determinant -1, not $+1$. Thus it lies in $\mathrm{GL}_2(\mathbb{Z})$, and a product with an even number of factors lies in $\mathrm{SL}_2(\mathbb{Z})$. More important than the sign is *unimodularity*: every product has determinant ± 1 and therefore has an inverse with integer entries. Appendix A develops the small amount of matrix algebra behind this observation. Appendix B then uses the same Möbius machinery to perform exact arithmetic directly on continued-fraction streams.

For example,

$$D(4)D(2)D(6)D(7) = \begin{pmatrix} 415 & 58 \\ 93 & 13 \end{pmatrix}, \qquad \begin{pmatrix} 415 & 58 \\ 93 & 13 \end{pmatrix}^{-1} = \begin{pmatrix} 13 & -58 \\ -93 & 415 \end{pmatrix}.$$

The entries of the inverse already display the certificate $1 = 13 \cdot 415 - 58 \cdot 93$. This is the first payoff of invertibility; the next section explains why it is exactly the extended Euclidean algorithm.

3.3 Extended Euclid in the convergents

The convergents do not merely reuse Euclid's quotients: their numerators and denominators are exactly the coefficients produced by the extended Euclidean algorithm.

Theorem 3.2 (Remainders from convergents)**.** *Let $A > B > 0$, and write the Euclidean divisions as*

$$r_{-2} = A, \qquad r_{-1} = B, \qquad r_{k-2} = a_k r_{k-1} + r_k \quad (k = 0, 1, \ldots, n),$$

where $r_{n-1} = \gcd(A, B)$ and $r_n = 0$. If

$$\frac{A}{B} = [a_0; a_1, \ldots, a_n] \quad \textit{and} \quad \frac{p_k}{q_k} = [a_0; a_1, \ldots, a_k],$$

then

$$\boxed{r_k = (-1)^k (q_k A - p_k B)} \qquad (0 \le k \le n).$$

In particular, the penultimate convergent gives a Bézout identity for the gcd.

Proof. For $k = 0$, the formula reads $r_0 = A - a_0 B$, the first Euclidean division. For $k = 1$,

$$r_1 = B - a_1 r_0 = -a_1 A + (a_0 a_1 + 1)B = -q_1 A + p_1 B.$$

If the formula holds for the preceding two remainders, then

$$\begin{aligned} r_k &= r_{k-2} - a_k r_{k-1} \\ &= (-1)^k((a_k q_{k-1} + q_{k-2})A - (a_k p_{k-1} + p_{k-2})B) \\ &= (-1)^k(q_k A - p_k B), \end{aligned}$$

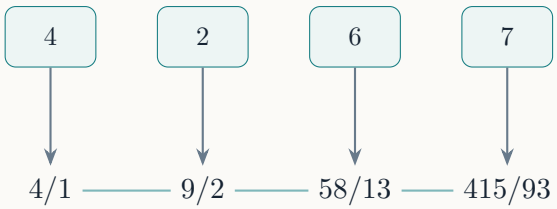
using the convergent recurrences. This proves the formula by induction. □

For $A/B = 415/93$, the two algorithms can be read in the same table:

k	a_k	convergent p_k/q_k	signed linear combination r_k
0	4	4/1	$415 - 4(93) = 43$
1	2	9/2	$-(2(415) - 9(93)) = 7$
2	6	58/13	$13(415) - 58(93) = 1$
3	7	415/93	$-(93(415) - 415(93)) = 0$

Thus the convergent just before 415/93 contains the certificate

$$1 = 13 \cdot 415 - 58 \cdot 93.$$



Each convergent remembers all earlier partial quotients.

Theorem 3.3 (Determinant identity)**.** *Consecutive convergents satisfy*

$$p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}.$$

Consequently, each p_k/q_k is in lowest terms and

$$\left| \frac{p_k}{q_k} - \frac{p_{k-1}}{q_{k-1}} \right| = \frac{1}{q_k q_{k-1}}.$$

Proof. Take determinants in (3.1). The determinant on the left is $(-1)^{k+1}$, while the determinant on the right is $p_k q_{k-1} - p_{k-1} q_k$. Since $(-1)^{k+1} = (-1)^{k-1}$, the identity follows. □

3.4 Infinite continued fractions

For an irrational real number x, repeatedly take integer parts:

$$a_0 = \lfloor x \rfloor, \qquad x_1 = \frac{1}{x - a_0}, \qquad a_1 = \lfloor x_1 \rfloor, \quad \dots$$

This produces $x = [a_0; a_1, a_2, \dots]$.

Example 3.4 (The silver staircase)**.** Let $x = \sqrt{2}$. Since

$$\frac{1}{\sqrt{2}-1} = \sqrt{2}+1 = 2+(\sqrt{2}-1),$$

the same remainder repeats, so

$$\sqrt{2} = [1;\overline{2}] = [1;2,2,2,\dots].$$

Its first convergents are

$$1,\quad \frac{3}{2},\quad \frac{7}{5},\quad \frac{17}{12},\quad \frac{41}{29},\dots$$

They alternately fall below and above $\sqrt{2}$.

Theorem 3.5 (The alternating squeeze)**.** *Let $c_k = p_k/q_k$ be the zero-indexed convergents of an irrational number $x = [a_0; a_1, a_2, \dots]$. Then*

$$c_0 < c_2 < c_4 < \cdots < x < \cdots < c_5 < c_3 < c_1.$$

Thus, when the convergents are counted starting with the first, the odd-numbered ones increase and stay below x, while the even-numbered ones decrease and stay above x.

The Möbius derivative trick. Fix k and leave the tail variable. The continued fraction with prefix through a_k is the fractional linear, or Möbius, map

$$M_k(t) = [a_0; a_1, \dots, a_k, t] = \frac{p_k t + p_{k-1}}{q_k t + q_{k-1}}.$$

The determinant identity gives

$$M_k'(t) = \frac{p_k q_{k-1} - p_{k-1} q_k}{(q_k t + q_{k-1})^2} = \frac{(-1)^{k+1}}{(q_k t + q_{k-1})^2}.$$

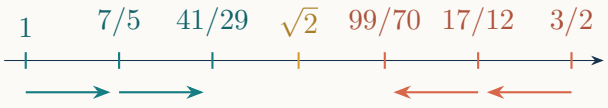
The complete tail $\alpha_{k+1} = [a_{k+1}; a_{k+2}, \dots]$ is finite and positive, and

$$x = M_k(\alpha_{k+1}), \qquad c_k = M_k(\infty).$$

When k is even, M_k is decreasing, so $c_k < x$; when k is odd, M_k is increasing, so $x < c_k$. The same comparison with $c_{k+2} = M_k([a_{k+1}; a_{k+2}])$ shows that $c_k < c_{k+2}$ for even k and $c_{k+2} < c_k$ for odd k. These are exactly the two monotone chains displayed above. □

Big idea

Every new convergent switches sides, but every second convergent moves inward. Continued fractions therefore trap x between a rising sequence of lower bounds and a falling sequence of upper bounds.



Schematic, not to scale: both parity subsequences move inward toward $\sqrt{2}$.

Theorem 3.6 (Approximation by convergents)**.** *If p_k/q_k is a convergent of an irrational x, then*

$$\left|x - \frac{p_k}{q_k}\right| < \frac{1}{q_k q_{k+1}} < \frac{1}{q_k^2}.$$

Moreover, if $0 < q < q_{k+1}$, then no fraction p/q is closer to x than both neighboring convergents in the appropriate interval. In this precise sense, convergents are best approximations with small denominators.

The determinant identity is the engine: distinct reduced fractions $a/b, c/d$ are at least $1/(bd)$ apart. There is simply not enough room for another small-denominator fraction between a convergent and x.

Theorem 3.7 (Legendre's approximation criterion)**.** *Let x be irrational and let p/q be in lowest terms with $q > 0$. If*

$$\left|x - \frac{p}{q}\right| < \frac{1}{2q^2},$$

then p/q is a convergent of x.

The theorem is a partial converse to the approximation theorem: an *exceptionally* good rational approximation cannot have arisen by accident. A standard proof chooses the two consecutive convergents whose denominators surround q and writes the integer vector (p, q) in their unimodular basis. If (p, q) is not one of the basis vectors, the determinant ± 1 forces it to be at least $1/(2q^2)$ away from x. This is another place where the integral inverse of the convergent matrix does real work.

Aside: continued fractions keep the calendar

A calendar must use whole days. Write the tropical year as $365 + \alpha$ days, where $\alpha \approx 0.2421988$. A q-year cycle with p leap days therefore asks for $p/q \approx \alpha$. Now

$$\alpha = [0; 4, 7, 1, 3, 5, \dots], \qquad \text{with convergents} \qquad \frac{1}{4}, \frac{7}{29}, \frac{8}{33}, \frac{31}{128}, \frac{163}{673}, \dots$$

The first convergent is the Julian rule: one leap day every four years. The convergent $31/128 = 0.2421875$ is remarkably accurate, but a 128-year civil cycle is awkward.
The Gregorian rule adds a practical constraint: organize corrections by centuries. Writing $q = 100Q$ changes the approximation problem:

$$q = 100Q: \qquad \frac{p}{Q} \approx 100\alpha \approx 24.21988, \qquad [24; 4] = \frac{97}{4}.$$

Thus the first nonintegral convergent of 100α gives 97 leap days in 400 years and the mean length

$$365 + \frac{97}{400} = 365.2425.$$

Operationally, the rule counts 100 multiples of 4, removes the four century years, and restores the one multiple of 400: $100 - 4 + 1 = 97$.
Notice that $97/400$ is not a convergent of α itself; it emerges after the century constraint. This is a mathematical reconstruction, not a claim about the reformers' historical method. It shows that "best" depends on which denominators an application permits.

Aside: continued fractions are roughly "base 11"

Khinchin's theorem says that for almost every real number,

$$(a_1 a_2 \cdots a_n)^{1/n} \longrightarrow K_0 \approx 2.685452,$$

the universal *Khinchin constant*. For efficiency, Lévy's companion theorem is more direct:

$$q_n^{1/n} \longrightarrow L = e^{\pi^2/(12\log 2)} \approx 3.275823.$$

Since convergent error is on the scale q_n^{-2}, it follows that

$$|x - p_n/q_n|^{-1/n} \longrightarrow L^2 = e^{\pi^2/(6\log 2)} \approx 10.731.$$

That is the "base 11" joke: one generic continued-fraction term buys about $\log_{10}(10.731) \approx 1.0306$ decimal digits. Lochs' theorem makes the comparison exact: N known decimal digits determine about $(6\log 2 \log 10/\pi^2)N \approx 0.97027N$ continued-fraction terms.
"Almost every" matters: a partial quotient is not literally a fixed-base digit, and quadratic surds such as $\sqrt{2}$ have exceptional periodic growth.

Experiment: π looks typical; e does not

Write

$$G_n(x) = (a_1 a_2 \cdots a_n)^{1/n}, \qquad B_n(x) = q_n^{2/n}.$$

Khinchin and Lévy predict, for almost every x, that $G_n(x) \to 2.685452\ldots$ and $B_n(x) \to 10.731016\ldots$. For

$$\pi = [3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, \ldots],$$

a computation gives the following rounded values:

n	$G_n(\pi)$	$B_n(\pi)$	$\log_{10} B_n(\pi)$
100	2.68315	10.68768	1.02888
1,000	2.66270	10.54434	1.02302
10,000	2.66320	10.57847	1.02442
100,000	2.67956	10.67835	1.02850
almost-everywhere target	2.68545	10.73102	1.03064

The wobble is real: rare large partial quotients, such as the early 292, produce visible jumps. A published computation using 17,001,303 terms found $G_n(\pi) = 2.686393$. This is compelling evidence, but no theorem is known that π satisfies either limiting law.
The number e is a useful contrast, not a second generic-looking example:

$$e = [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, \ldots].$$

For $n = 3m$, its first n noninitial partial quotients have product $2^m m!$. Stirling's formula therefore gives

$$G_{3m}(e) = (2^m m!)^{1/(3m)} \sim (2m/e)^{1/3} \longrightarrow \infty.$$

Also $q_{3m} \geq a_1 \cdots a_{3m} = 2^m m!$, so $B_{3m}(e) \to \infty$ as well. Thus e is provably exceptional for both laws, while π is numerically persuasive but theoretically mysterious.

Common trap

Legendre, Lagrange, and Galois all appear in this subject. Legendre's theorem recognizes very good rational approximations. Lagrange's theorem characterizes eventually periodic continued fractions of quadratic irrationals; Galois's refinement says exactly when the period begins immediately.

3.5 A parallel periodicity: digits in base b

Continued fractions are not the first place students meet repeating expansions. Let $0 < m < q$ and write $x = m/q$ in base $b \geq 2$. Long division produces digits $d_1, d_2, \ldots$ and remainders $r_0, r_1, \ldots$ by

$$r_0 = m, \qquad br_n = qd_{n+1} + r_{n+1}, qquad 0 \leq r_{n+1} < q.$$

There are only q possible remainders. If a remainder becomes 0, the expansion terminates; otherwise some remainder repeats, and from that point the deterministic long-division process repeats. Thus every rational number has an eventually periodic base-b expansion.

The converse uses only a geometric series. If a block of ℓ digits with integer value N repeats immediately, then

$$0.\overline{d_1 \cdots d_\ell}_b = \frac{N}{b^\ell} + \frac{N}{b^{2\ell}} + \cdots = \frac{N}{b^\ell - 1}.$$

An initial nonrepeating block only adds a rational number with denominator a power of b.

Theorem 3.8 (When the repeating block begins immediately)**.** *If the reduced fraction m/q has a purely periodic base-b expansion, then* $\gcd(q, b) = 1$*. Conversely, if* $\gcd(q, b) = 1$*, its expansion is purely periodic.*

The direction available now. A purely periodic expansion has the form $N/(b^\ell - 1)$. Its reduced denominator divides $b^\ell - 1$, which is relatively prime to b. □

The reverse implication is deliberately postponed until Chapter 6. Euler's theorem will give

$$b^{\varphi(q)} \equiv 1 \pmod{q},$$

so $q \mid b^{\varphi(q)} - 1$; rewriting m/q with that denominator creates a repeating block (possibly with leading zeros). In decimal, for example,

$$\frac{1}{7} = 0.\overline{142857}, \qquad \frac{1}{8} = 0.125000\ldots.$$

The first expansion is purely periodic because $\gcd(7, 10) = 1$; the second is only eventually periodic.

3.6 Preview: quadratic surds repeat

Definition 3.9. A *quadratic irrational* (or quadratic surd) is an irrational real root of a quadratic polynomial with integer coefficients. Its other root is its *conjugate*, denoted x'.

Theorem 3.10 (Lagrange)**.** *A real number has an eventually periodic simple continued fraction if and only if it is a quadratic irrational.*

The easier direction is now visible from matrices. If a block of digits repeats, the corresponding product $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ sends the repeating tail t back to itself:

$$t = \frac{at + b}{ct + d}.$$

Consequently $ct^2 + (d - a)t - b = 0$. Adding a finite prefix applies one more integer Möbius transformation, so the original number is also quadratic.

For the deeper direction, one writes each complete quotient of a quadratic irrational in the form

$$x_n = \frac{P_n + \sqrt{D}}{Q_n}.$$

The continued-fraction step gives integer recurrences for P_n, Q_n. After a short initial segment, inequalities involving x_n and its conjugate bound both integers. Only finitely many states (P_n, Q_n) remain, so one repeats; the deterministic algorithm is periodic thereafter. We will use this as a preview theorem rather than prove all of the bounding details here.

Definition 3.11. A quadratic irrational x is *reduced* if

$$x > 1 \qquad \text{and} \qquad -1 < x' < 0.$$

Theorem 3.12 (Galois's pure-periodicity criterion)**.** *The simple continued fraction of a quadratic irrational x is purely periodic if and only if x is reduced.*

The conjugate inequalities describe exactly when the continued-fraction algorithm stays inside the finite collection of reduced surds from its first step, rather than entering it after a transient prefix. Thus

$$\sqrt{2} = [1; \overline{2}]$$

is eventually but not purely periodic because its conjugate $-\sqrt{2}$ is less than -1, while

$$1 + \sqrt{2} = [\overline{2}]$$

is purely periodic because $1 - \sqrt{2}$ lies between -1 and 0.

3.7 Periodicity meets Pell's equation

The preceding theorems and Pell's equation fit together in both directions. Suppose $D \geq 2$ is not a square and

$$p^2 - Dq^2 = \pm 1, \qquad p, q > 0.$$

Then

$$\left|\sqrt{D} - \frac{p}{q}\right| = \frac{1}{q^2(\sqrt{D} + p/q)} < \frac{1}{2q^2}. \tag{3.2}$$

Legendre's criterion therefore says that p/q must be a convergent of $\sqrt{D}$. Pell solutions are not merely good approximations; continued fractions are forced to find them.

In the other direction, Lagrange says

$$\sqrt{D} = [a_0; \overline{a_1, \ldots, a_L}].$$

Galois explains why the repeating part starts after a_0: the first complete quotient $1/(\sqrt{D}-a_0)$ is reduced. A classical consequence of the matrix recurrence is

$$p_{L-1}^2 - Dq_{L-1}^2 = (-1)^L.$$

Thus one trip through an even-length period gives a solution of the $+1$ equation. If L is odd, one trip gives -1 and two trips give $+1$.

Big idea

$$\text{periodic surd} \implies \text{period matrix} \implies \text{Pell solutions,}$$

$$\text{Pell solution} \implies \text{error below } 1/(2q^2) \xRightarrow{\text{Legendre}} \text{convergent.}$$

Periodicity produces the solutions, and exceptional approximation recognizes them.

For the convergents p/q to $\sqrt{2}$, direct computation gives

$$p^2 - 2q^2 = \pm 1.$$

Here the period has odd length 1: $1/1$ solves the -1 equation, while two copies of the period produce $3/2$ and the $+1$ equation. The positive solutions to $p^2 - 2q^2 = 1$ begin $(3,2), (17,12), (99,70)$.

There is also a direct matrix generator. Whenever $u^2 - Dv^2 = 1$,

$$U = \begin{pmatrix} u & Dv \\ v & u \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

If $(p,q)^T$ represents $p + q\sqrt{D}$, then $U(p,q)^T$ represents $(u + v\sqrt{D})(p + q\sqrt{D})$. Hence powers of U generate new Pell solutions. For $D = 2$ the fundamental solution $(u,v) = (3,2)$ gives

$$\begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}^n \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} p_n \\ q_n \end{pmatrix}, \qquad p_n^2 - 2q_n^2 = 1.$$

Checkpoint

Compute $99/70$ as a decimal. The striking accuracy is not a floating-point accident: $99^2 - 2 \cdot 70^2 = 1$ forces

$$\left|\sqrt{2} - 99/70\right| = \frac{1}{70^2(\sqrt{2} + 99/70)}.$$

Exercises

1. Find the continued fraction of $355/113$. List all convergents. [Fluency]
2. Compute the first six convergents of $[1;\overline{1}]$. What familiar sequence appears? [Fluency]
3. Prove the determinant identity by induction. Use it to prove $\gcd(p_k, q_k) = 1$. [Reasoning]
4. Show that $[a_0;\ldots,a_n] = [a_0;\ldots,a_n - 1, 1]$ when $a_n > 1$. Explain the only ambiguity in finite expansions. [Reasoning]
5. Derive $\sqrt{3} = [1;\overline{1,2}]$ and compute four convergents. [Reasoning]

6. Write the tropical year as $365 + \alpha$ with $\alpha \approx 0.2421988$. Compute the first four nonzero convergents of α and identify the Julian rule. Then impose a century-based cycle by approximating 100α; explain how $97/4$ becomes the Gregorian rule $97/400$. Compare it with $31/128$. [Exploration]

7. Investigate which convergents to $\sqrt{2}$ solve $p^2 - 2q^2 = 1$ and which solve -1. State and prove a pattern using the recurrence. [Challenge]

8. Let $A/B = [a_0; \ldots, a_n]$ and use the convergent recurrences to prove $r_k = (-1)^k(q_kA - p_kB)$. Apply the penultimate convergent of $527/341$ to recover a Bézout identity for $\gcd(527, 341)$. [Reasoning]

9. For $x = \sqrt{3}$, compute enough convergents to display three lower bounds and three upper bounds. Verify that the lower bounds increase and the upper bounds decrease, then explain why the Möbius derivative has the sign needed for each chain. [Reasoning]

10. Prove the matrix product formula (3.1) by induction. Explain why its inverse always has integer entries. [Reasoning]

11. Long-divide $1/7$ and $5/12$ in base 10, recording the remainder at every step. Use repeated remainders to predict the preperiod and period. [Fluency]

12. Prove directly from a geometric series that every eventually periodic base-b expansion is rational. [Reasoning]

13. After studying Euler's theorem, return to prove that $\gcd(q, b) = 1$ implies the base-b expansion of m/q is purely periodic. Give an explicit upper bound for a period length. [Reasoning]

14. Verify that $355/113$ satisfies Legendre's inequality for π using a calculator value of π. What conclusion does the theorem certify? [Exploration]

15. Let $t = [\overline{1, 2}]$. Use the matrix $D(1)D(2)$ to find a quadratic equation for t, and identify the positive root. [Reasoning]

16. Check Galois's inequalities for $1 + \sqrt{2}$ and $(1 + \sqrt{5})/2$. Explain why their continued fractions are purely periodic. [Reasoning]

17. Let $p^2 - Dq^2 = \pm 1$ with $p, q > 0$. Derive (3.2), verify its strict Legendre bound, and conclude that p/q is a convergent of $\sqrt{D}$. [Reasoning]

18. Compute the period of $\sqrt{3}$. Use its length to predict whether one period solves the $+1$ or -1 Pell equation. Then use powers of $\left(\begin{smallmatrix} 2 & 3 \\ 1 & 2 \end{smallmatrix}\right)$ to generate three solutions of $p^2 - 3q^2 = 1$. [Exploration]

19. Define $G_n(x) = (a_1 \cdots a_n)^{1/n}$ and $B_n(x) = q_n^{2/n}$. Use the Sage notebook to reproduce several rows of the π table above. Explain why agreement with the Khinchin–Lévy constants is evidence but not proof. Then derive the displayed formula for $G_{3m}(e)$ directly from the pattern in the continued fraction of e. [Exploration]

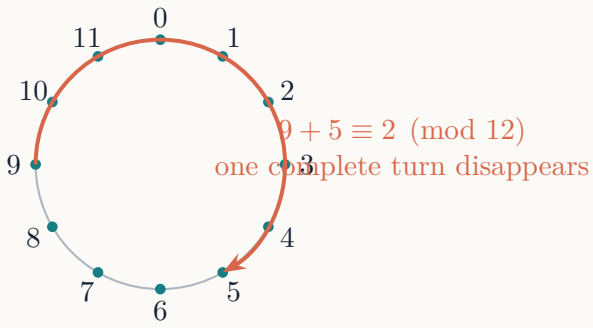
Chapter 4

Arithmetic on a clock

Congruences, inverses, and the Chinese remainder theorem

Definition 4.1. For $n \geq 2$, we write $a \equiv b \pmod{n}$ if $n \mid (a - b)$. The equivalence class of a is written $[a]$ or $\bar{a}$.

Congruence remembers the remainder and forgets the number of complete turns around the clock.



4.1 Legal arithmetic

If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

$$a + c \equiv b + d \pmod{n}, \qquad ac \equiv bd \pmod{n}.$$

Powers are therefore legal too. Division is different.

Definition 4.2. A residue a is a *unit modulo* n if there exists b with $ab \equiv 1 \pmod{n}$. Such a b is an inverse of a.

Theorem 4.3 (Invertibility criterion)**.** *a has an inverse modulo n exactly when* $\gcd(a, n) = 1$.

Proof. The congruence $ab \equiv 1 \pmod{n}$ means $ab + nk = 1$ for some k. This is precisely a Bézout identity. Such an identity exists exactly when $\gcd(a, n) = 1$. □

Common trap

Cancellation can fail. Modulo 6, $2 \cdot 1 \equiv 2 \cdot 4$, but $1 \not\equiv 4$. You may cancel a modulo n only when a is a unit, or after carefully reducing the modulus.

Proposition 4.4 (Linear congruence)**.** *The congruence $ax \equiv b \pmod{n}$ has a solution exactly when $d = \gcd(a, n)$ divides b. When it does, there are exactly d incongruent solutions modulo n.*

4.2 Reassembling remainders

Suppose a number leaves remainder 2 modulo 3 and remainder 3 modulo 5. Listing possibilities gives 8 modulo 15. The Chinese remainder theorem says this always works for coprime clocks.

Theorem 4.5 (Chinese remainder theorem)**.** *If $n_1, \dots, n_r$ are pairwise coprime, then for any $a_1, \dots, a_r$, the system*

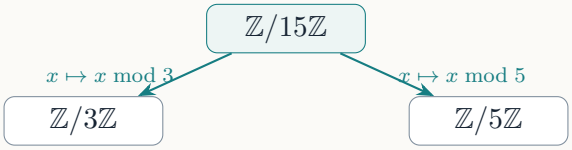
$$x \equiv a_i \pmod{n_i} \qquad (i = 1, \dots, r)$$

has a solution, unique modulo $N = n_1 \cdots n_r$.

Construction. Set $N_i = N/n_i$. Since $\gcd(N_i, n_i) = 1$, choose M_i with $N_i M_i \equiv 1 \pmod{n_i}$. Then

$$x = \sum_{i=1}^{r} a_i N_i M_i$$

has every prescribed remainder: modulo n_i, its ith term is a_i and every other term is 0. If x, y both work, each n_i divides $x - y$; coprimality makes $N \mid x - y$. □



One clock of size 15 is equivalent to two independent clocks.

4.3 Fast powers and patterns

Modular reduction prevents numbers from exploding. To compute $7^{222} \pmod{13}$, square and reduce:

$$7^2 \equiv 10, \quad 7^4 \equiv 9, \quad 7^8 \equiv 3, \quad 7^{16} \equiv 9 \pmod{13},$$

then use the binary expansion $222 = 128 + 64 + 16 + 8 + 4 + 2$. Later, Fermat's theorem will make this even faster.

Explore

Make a multiplication table modulo n and highlight the rows that are permutations of all residues. Conjecture which row labels have inverses. Then compare with the gcd criterion.

Exercises

1. Find the inverse of 37 modulo 101 using extended Euclid. [Fluency]

2. Solve $18x \equiv 30 \pmod{42}$ and list all solutions modulo 42. [Fluency]

3. Prove that congruence modulo n is an equivalence relation. [Reasoning]

4. Show that an integer is divisible by 9 exactly when its decimal digit sum is divisible by 9. [Reasoning]

5. Solve $x \equiv 2 \pmod{7}$, $x \equiv 3 \pmod{8}$, $x \equiv 4 \pmod{9}$ by the constructive CRT proof. [Reasoning]

6. Determine every possible last two digits of a square. Explain how CRT can split the work into moduli 4 and 25. [Exploration]

7. Prove that $x^2 \equiv 1 \pmod{p}$ has only the solutions $x \equiv \pm 1$ when p is prime. Find a composite modulus where there are more than two solutions. [Challenge]

Chapter 5

A theorem you can rotate

Necklace colorings and Fermat's little theorem

Why should a prime p divide $a^p - a$ for every integer a? Instead of manipulating the expression, we count objects that naturally arrive in packets of p.

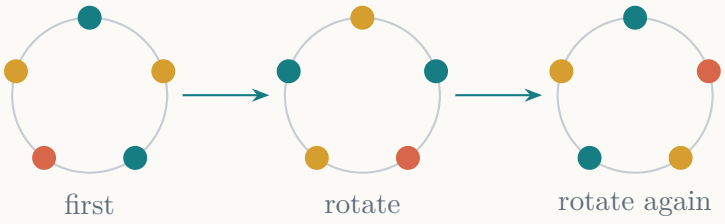
5.1 Words on a circular loom

Fix a prime p and a colors. First imagine p labeled positions around a circle. A coloring is a word

$$(c_0, c_1, \ldots, c_{p-1}),$$

and there are a^p such words. Rotation by one step sends it to

$$(c_{p-1}, c_0, \ldots, c_{p-2}).$$



Group words together when one is a rotation of another. Such a group is called a rotational *orbit*. How large can an orbit be?

Lemma 5.1 (Prime orbit dichotomy)**.** *For p prime, a length-p word has either 1 or p distinct rotations. It has one exactly when it is constant.*

Proof. If rotation by k positions, with $1 \le k < p$, fixes the word, then entries i and $i + k$ have the same color (indices modulo p). Since $\gcd(k, p) = 1$, repeatedly adding k visits every position. Thus all entries have one color. A nonconstant word has no repeated rotation before the pth, so its orbit has size p. □

There are a constant words, one for each color. The remaining $a^p - a$ words split into orbits of exactly p words.

Theorem 5.2 (Fermat's little theorem, necklace form)**.** *If p is prime, then for every integer a,*

$$a^p \equiv a \pmod{p}.$$

If $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Proof. For $a \geq 0$, the orbit count above shows $p \mid a^p - a$. This congruence depends only on the residue of a modulo p, so it holds for every integer a. If $p \nmid a$, then a has an inverse modulo p, and we may cancel it from $a^p \equiv a$. □

Big idea

Divisibility by p appears because symmetry packages nonexceptional objects into groups of p. The exceptions are exactly the colorings fixed by rotation. This pattern – count, isolate fixed points, divide the rest into orbits – reaches far beyond number theory.

5.2 A subtle choice: labeled words or necklaces?

A physical necklace usually ignores where position 0 is. Here we first count *labeled* circular words, then use rotation to partition them. Calling each orbit a necklace is convenient, but the proof depends on keeping those two levels distinct.

Common trap

When the length is composite, an orbit can have an intermediate size. The word $RBRB$ has two distinct rotations, not 1 or 4. Primality is doing real work.

5.3 Using the theorem

To compute $3^{1000} \pmod{7}$, note that $3^6 \equiv 1$ and $1000 = 6 \cdot 166 + 4$:

$$3^{1000} \equiv (3^6)^{166} 3^4 \equiv 81 \equiv 4 \pmod{7}.$$

Fermat's theorem also explains the first layer of binomial divisibility. For $1 \leq k \leq p-1$,

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}$$

is divisible by p, because the denominator contains no factor p. Therefore

$$(x+y)^p \equiv x^p + y^p \pmod{p}.$$

We will return to this "freshman's dream" after developing binomial coefficients systematically.

Exercises

1. List the rotational orbits of binary words of length 3. Identify the constant words. [Fluency]

2. Compute $11^{2026} \pmod{17}$ using Fermat's theorem. [Fluency]

3. In the orbit lemma, prove carefully that $0, k, 2k, \ldots, (p-1)k$ are all distinct modulo p. [Reasoning]

4. Give an orbit-counting proof that $p \mid a^p - a$ using linear strings and cyclic shift, without drawing a circle. [Reasoning]

5. Find every possible orbit size for binary words of length 6, and give an example of each. [Exploration]

6. Prove that if p is prime, then $(x+y)^p \equiv x^p + y^p \pmod{p}$. Use induction to deduce $(x_1 + \cdots + x_m)^p \equiv x_1^p + \cdots + x_m^p$. [Reasoning]

7. If n is composite, must there be an a with $a^n \not\equiv a \pmod{n}$? Test $n = 341$ and 561. This opens the door to pseudoprimes and Carmichael numbers. [Challenge]

Chapter 6

The reusable machine

A quick introduction to finite groups

The necklace proof is vivid, but it was tailored to one symmetry. Group theory isolates the small collection of rules shared by rotations, invertible residues, permutations, and many other systems.

6.1 One operation, four rules

Definition 6.1. A *group* is a set G with an operation $*$ such that:

1. $a * b \in G$ for all $a, b \in G$ (closure);

2. $(a * b) * c = a * (b * c)$ (associativity);

3. there is an identity e with $e * a = a * e = a$;

4. each a has an inverse a^{-1} with $a * a^{-1} = a^{-1} * a = e$.

If also $a * b = b * a$, the group is *abelian*.

Group	Operation	Identity / inverse
$\mathbb{Z}/n\mathbb{Z}$	addition	0 / $-a$
$(\mathbb{Z}/n\mathbb{Z})^\times$	multiplication of units	1 / modular inverse
rotations of a regular n-gon	composition	no rotation / reverse rotation
permutations of a set	composition	do nothing / undo

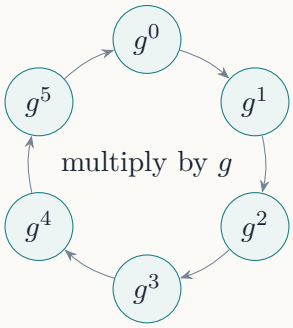
Not every familiar set works: all residues modulo n under multiplication fail because nonunits have no inverse.

6.2 Powers and cyclic subgroups

For $g \in G$, the list

$$e, g, g^2, g^3, \dots$$

eventually repeats when G is finite. The first positive m with $g^m = e$ is the *order* of g, written $\operatorname{ord}(g)$. The powers form the cyclic subgroup $\langle g \rangle$.



A cyclic group of order 6: repeated multiplication traces a loop.

6.3 Cosets and Lagrange's theorem

If H is a subgroup of G and $g \in G$, the left coset

$$gH = \{gh : h \in H\}$$

is a translated copy of H. Two cosets are either equal or disjoint, and every coset has $|H|$ elements because $h \mapsto gh$ is a bijection.

Theorem 6.2 (Lagrange)**.** *If G is finite and $H \leq G$, then $|H|$ divides $|G|$.*

Proof. The distinct cosets of H partition G into equally sized blocks. If there are r blocks, then $|G| = r\,|H|$. □

Corollary 6.3. *For every g in a finite group G, $\operatorname{ord}(g) \mid |G|$ and hence $g^{|G|} = e$.*

6.4 Fermat's theorem, now in one line

For prime p, every nonzero residue has an inverse, so

$$(\mathbb{Z}/p\mathbb{Z})^{\times} = \{1, 2, \ldots, p-1\}$$

is a group of order $p - 1$. Lagrange immediately yields the standard group-theoretic proof:

$$a^{p-1} \equiv 1 \pmod{p} \qquad (p \nmid a).$$

The gain is not a shorter proof at any price. The gain is a theorem that can be reused.

Theorem 6.4 (Euler)**.** *If $\gcd(a, n) = 1$, then*

$$a^{\varphi(n)} \equiv 1 \pmod{n},$$

where $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^{\times}|$.

Proof. Apply the corollary of Lagrange to the group $(\mathbb{Z}/n\mathbb{Z})^{\times}$. □

Big idea

The necklace proof uses orbits of a cyclic action; Lagrange uses cosets of a subgroup. Both prove divisibility by partitioning a finite set into equal blocks. Group theory turns that shared move into a portable language.

6.5 A second standard proof without naming Lagrange

If $a \not\equiv 0 \pmod{p}$, multiplication by a permutes the nonzero residues:

$$a, 2a, \ldots, (p-1)a \pmod{p}$$

is just $1, 2, \ldots, p-1$ in another order. Multiply all entries:

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Since $(p-1)!$ is invertible, cancel it. This is the group proof with the machinery folded away.

Exercises

1. List $(\mathbb{Z}/12\mathbb{Z})^\times$ and make its multiplication table. Find the order of every element. [Fluency]
2. Find $\varphi(20)$ and use Euler's theorem to reduce 3^{2026} modulo 20. [Fluency]
3. Prove that the identity of a group is unique and that each element has a unique inverse. [Reasoning]
4. Prove that two left cosets of H are equal or disjoint. [Reasoning]
5. Show that a group of prime order is cyclic. [Reasoning]
6. Draw the subgroup lattice of $(\mathbb{Z}/12\mathbb{Z}, +)$. Check Lagrange's theorem at every node. [Exploration]
7. Prove $\varphi(mn) = \varphi(m)\varphi(n)$ when $\gcd(m, n) = 1$ by using the Chinese remainder theorem. [Challenge]
8. Compare the necklace, permutation, and Lagrange proofs of Fermat's theorem. For each, identify the set being partitioned or permuted and the exact place primality enters. [Synthesis]

Chapter 7

Arithmetic in Pascal's triangle

Binomial coefficients and divisibility

Definition 7.1. For $0 \le k \le n$,

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

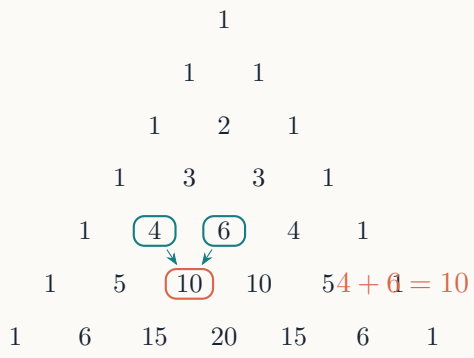
It counts the k-element subsets of an n-element set.

The counting meaning proves integrality without asking the fraction to look integral. It also makes identities visible.

Proposition 7.2 (Pascal and symmetry)**.**

$$\binom{n}{k} = \binom{n}{n-k}, \qquad \binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

For Pascal's identity, distinguish k-subsets that contain a chosen element from those that do not.



7.1 Three ways to see the binomial theorem

Theorem 7.3 (Binomial theorem)**.**

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

In expanding $(x+y)^n$, choose which k of the n factors contribute y. This proof also explains why the coefficients are integers and why their sum is 2^n.

Proposition 7.4 (Vandermonde)**.** *For nonnegative integers r, s, n,*

$$\sum_k \binom{r}{k}\binom{s}{n-k} = \binom{r+s}{n}.$$

Count n-person committees from two groups: either all at once, or according to how many come from the first group.

7.2 Prime rows

Theorem 7.5. *If p is prime and $1 \le k \le p-1$, then*

$$p \mid \binom{p}{k}.$$

Proof. The identity $k\binom{p}{k} = p\binom{p-1}{k-1}$ shows p divides $k\binom{p}{k}$. Since $p \nmid k$, Euclid's lemma gives $p \mid \binom{p}{k}$. □

Thus the pth row of Pascal's triangle becomes $1, 0, \ldots, 0, 1$ modulo p, which again yields

$$(x+y)^p \equiv x^p + y^p \pmod{p}.$$

7.3 Measuring divisibility one prime at a time

Definition 7.6. For nonzero integer m, $v_p(m)$ is the exponent of p in its prime factorization. Thus $v_2(40) = 3$.

Valuations turn multiplication into addition:

$$v_p(ab) = v_p(a) + v_p(b), \qquad v_p\left(\frac{a}{b}\right) = v_p(a) - v_p(b)$$

when the quotient is an integer.

Theorem 7.7 (Legendre)**.**

$$v_p(n!) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots.$$

Proof. Among $1, \ldots, n$, exactly $\lfloor n/p \rfloor$ contribute at least one p; $\lfloor n/p^2 \rfloor$ contribute one additional p; and so on. □

Consequently,

$$v_p\binom{n}{k} = \sum_{j \ge 1} \left(\left\lfloor \frac{n}{p^j} \right\rfloor - \left\lfloor \frac{k}{p^j} \right\rfloor - \left\lfloor \frac{n-k}{p^j} \right\rfloor \right).$$

Each parenthesis is 0 or 1: it records whether adding k and $n-k$ creates a carry at the p^j place.

Theorem 7.8 (Kummer's carry rule)**.** *$v_p\binom{n}{k}$ equals the number of carries when k and $n-k$ are added in base p.*

The formula above is already the proof once carries are read from right to left. It gives a remarkable visual test: $\binom{n}{k}$ is not divisible by p precisely when subtracting k from n in base p needs no borrowing.

Example 7.9. To find $v_2\binom{10}{3}$, add $3 = 0011_2$ and $7 = 0111_2$:

$$\begin{array}{r} 0011 \\ +0111 \\ \hline 1010 \end{array}$$

There are three carries, so $v_2\binom{10}{3} = 3$. Indeed, $\binom{10}{3} = 120 = 2^3 \cdot 15$.

Explore

Color entry $\binom{n}{k}$ when it is nonzero modulo 2, then modulo 3. Zoom out. The self-similar patterns are the visual shadow of base-p digits and lead to Lucas's theorem.

Exercises

1. Prove $\sum_k \binom{n}{k} = 2^n$ and $\sum_k (-1)^k \binom{n}{k} = 0$ for $n > 0$. [Fluency]

2. Compute $v_2(100!)$, $v_5(100!)$, and the number of trailing zeros of $100!$. [Fluency]

3. Give a counting proof of $k\binom{n}{k} = n\binom{n-1}{k-1}$. [Reasoning]

4. Use Vandermonde's identity to prove $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$. [Reasoning]

5. Determine all n for which every interior entry of row n is even. Use Kummer's rule. [Reasoning]

6. For a fixed prime p, explore the first p^2 rows of Pascal's triangle modulo p. State a digit-based conjecture for when $\binom{n}{k} \not\equiv 0 \pmod{p}$. [Exploration]

7. Prove that $\operatorname{lcm}(1, 2, \ldots, 2n)$ is divisible by $\binom{2n}{n}$ by comparing p-adic valuations. [Bridge]

Chapter 8

How crowded are the primes?

Chebyshev's estimate from the middle of Pascal's triangle

Euclid proved that the primes never end. A sharper question asks how many lie below x:

$$\pi(x) = \#\{p \le x : p \text{ prime}\}.$$

The prime number theorem says $\pi(x) \sim x/\log x$, but its proof lies beyond this course. Chebyshev's elementary argument already captures the correct scale.

8.1 The central binomial squeeze

The largest coefficient in $(1+1)^{2n}$ is central. Since there are $2n+1$ nonnegative coefficients with sum 4^n,

$$\frac{4^n}{2n+1} \le \binom{2n}{n} < 4^n. \tag{8.1}$$

This elementary inequality contains surprising information about primes.

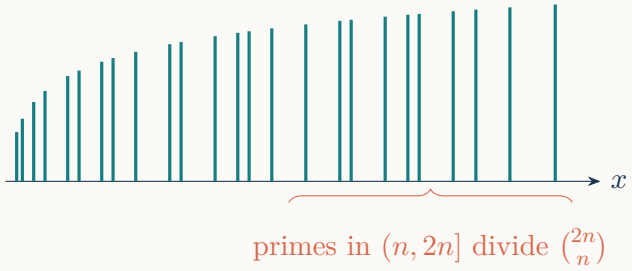
Define two weighted prime counts:

$$\vartheta(x) = \sum_{p \le x} \log p, \qquad \psi(x) = \sum_{p^k \le x} \log p.$$

The first logs the product of primes at most x; the second logs the least common multiple:

$$\psi(x) = \log \operatorname{lcm}(1, 2, \dots, \lfloor x \rfloor).$$

A logarithmic weight turns products into sums.



8.2 An upper bound for the prime product

Every prime $n < p \le 2n$ occurs in the numerator $(n+1)(n+2)\cdots(2n)$ and not in $n!$. Therefore it divides $\binom{2n}{n}$, and

$$\vartheta(2n) - \vartheta(n) \le \log \binom{2n}{n} < 2n \log 2.$$

Adding these inequalities over dyadic intervals gives the following deliberately non-optimized bound.

Lemma 8.1. *For $x \ge 2$,*

$$\vartheta(x) \le 2(\log 2)x.$$

Proof. Split $(1, x]$ into $(x/2, x], (x/4, x/2], \ldots$ and apply the preceding estimate, with integer endpoints rounded outward. The resulting geometric series is bounded by

$$x \log 2 + \frac{x}{2} \log 2 + \frac{x}{4} \log 2 + \cdots = 2x \log 2.$$

Endpoint rounding can only add primes already included in the neighboring interval. □

8.3 A lower bound from an lcm

The bridge exercise in the last chapter showed

$$\binom{2n}{n} \mid \operatorname{lcm}(1, 2, \ldots, 2n).$$

Taking logarithms and using (8.1),

$$\psi(2n) \ge \log \binom{2n}{n} \ge 2n \log 2 - \log(2n+1). \tag{8.2}$$

Prime powers make ψ larger than ϑ, but not by too much:

$$\psi(x) = \vartheta(x) + \vartheta(x^{1/2}) + \vartheta(x^{1/3}) + \cdots.$$

By Lemma 8.1, all terms after the first contribute at most

$$2\sqrt{x} \log x$$

for $x \ge 4$: there are at most $\log_2 x$ terms, and every argument is at most $\sqrt{x}$. Combining this with (8.2) shows that, once x is sufficiently large,

$$\vartheta(x) \ge \frac{\log 2}{2} x. \tag{8.3}$$

The constant $1/2$ is chosen for convenience; better bookkeeping improves it.

8.4 From weighted primes to ordinary counting

The lower bound is immediate because every term in $\vartheta(x)$ is at most $\log x$:

$$\vartheta(x) \le \pi(x)\log x \quad \Longrightarrow \quad \pi(x) \ge \frac{\log 2}{2}\frac{x}{\log x}.$$

For the upper bound, primes between $\sqrt{x}$ and x each contribute at least $\frac{1}{2}\log x$:

$$\vartheta(x) \ge (\pi(x) - \pi(\sqrt{x}))\frac{\log x}{2}.$$

Using $\pi(\sqrt{x}) \le \sqrt{x}$ and Lemma 8.1 gives

$$\pi(x) \le 4\log 2\,\frac{x}{\log x} + \sqrt{x} \le 4\frac{x}{\log x}$$

for all sufficiently large x.

Theorem 8.2 (Chebyshev estimate)**.** *For all sufficiently large x,*

$$\boxed{\frac{\log 2}{2}\frac{x}{\log x} \le \pi(x) \le 4\frac{x}{\log x}}$$

In particular, $\pi(x) \asymp x/\log x$: their ratio stays between two positive constants.

Common trap

The symbol $\asymp$ does not mean $\sim$. Chebyshev's estimate gets the correct order of magnitude. The prime number theorem makes the much stronger claim

$$\lim_{x\to\infty} \frac{\pi(x)}{x/\log x} = 1.$$

Big idea

Why did binomial coefficients see primes? Factorials contain every integer up to a boundary. Dividing $(2n)!$ by two copies of $n!$ cancels most small-scale information while exposing primes that cross the boundary from n to $2n$. The central coefficient is both arithmetically rich and analytically easy to estimate.

Exercises

1. Compute $\pi(10^k)$ and $10^k/\log(10^k)$ for $k = 1, 2, 3, 4$. Track the ratio. [Fluency]
2. Prove both sides of the central binomial squeeze (8.1). [Reasoning]
3. Prove that every prime p with $n < p \le 2n$ divides $\binom{2n}{n}$ exactly once. [Reasoning]
4. Fill in the valuation proof that $\binom{2n}{n}$ divides $\operatorname{lcm}(1, \ldots, 2n)$. [Reasoning]
5. Starting from the dyadic inequality, write a fully detailed proof of Lemma 8.1, including floors and ceilings. [Challenge]

6. Use a computer to graph $\pi(x)$, $x/\log x$, and the two Chebyshev bounds. Which statement is visible at moderate x, and which remains too coarse? [Exploration]

7. Show that Chebyshev's estimate implies the nth prime p_n has order of magnitude $n \log n$. [Synthesis]

Chapter 9

Capstone studio

Choose a thread, extend it, and explain it

The capstone is not a miniature research paper. It is a carefully explained mathematical journey: a motivating question, small examples, one substantial proof, and a clear account of how course ideas connect.

9.1 Project menu

A. Rational approximation and Pell equations

Choose a nonsquare d. Compute the continued fraction of $\sqrt{d}$, identify convergents that solve $x^2 - dy^2 = \pm 1$, and prove at least one recurring pattern. Include a number-line or lattice illustration.

B. Symmetry, orbits, and divisibility

Generalize necklace counting to length n. Classify possible orbit sizes for one composite n, explain stabilizers in concrete language, and derive a divisibility identity. Möbius inversion is an optional extension, not a prerequisite.

C. Pascal's triangle modulo a prime

Visualize Pascal's triangle modulo p. Develop and prove a digit rule (Lucas's theorem is the natural destination), then explain how Kummer's carry rule and the visual pattern encode the same arithmetic.

D. Prime-counting laboratory

Compare $\pi(x)$ with $x/\log x$, $\mathrm{Li}(x)$, and the Chebyshev bounds over a range your computer can handle. Explain which plots are evidence and which claims are proved. Reconstruct one link in Chebyshev's proof in full detail.

9.2 Required deliverables

1. A four-to-six page mathematical exposition.
2. At least one original diagram or data visualization.
3. One theorem stated precisely and proved completely.
4. A one-page "proof map" naming the definitions and earlier results used.
5. A five-minute oral explanation, followed by questions.

9.3 Evaluation rubric

Criterion	Evidence of strong work	Weight
Mathematical correctness	Claims are true, hypotheses are visible, examples are checked, and the main proof closes every logical gap.	35%
Exposition	The question motivates the definitions; notation is stable; paragraphs explain why steps matter.	25%
Connection	At least two course strands are linked in a mathematically meaningful way.	15%
Visual or computational insight	A diagram or computation reveals structure and is interpreted rather than merely displayed.	15%
Revision and response	The final version addresses peer/instructor feedback and the oral defense shows ownership.	10%

Checkpoint

Before submitting, underline every sentence that makes a mathematical claim. For each underline, ask: Is it a definition, an observed example, a cited theorem, or a claim I have proved? Ambiguous status is often the real source of unclear exposition.

9.4 A final synthesis prompt

In 500–700 words, respond to this statement:

> Number theory repeatedly converts multiplication into something easier to organize: remainders, exponent vectors, additive logarithms, or equal-sized orbits.

Use one precise example from three different chapters. Your response should include at least one equation and explain not only what the conversion does, but why it helps a proof.

Appendix A

Two-by-two matrices and Möbius maps

A compact toolkit for continued fractions

This appendix develops only the matrix algebra used in Chapter 3. The central idea is simple: a matrix with determinant ± 1 can be reversed without ever leaving the integers.

A.1 Multiplication and determinants

A 2×2 matrix acts on a column vector by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

Matrix multiplication is defined so that composition works as expected: applying B and then A is the same as applying AB. Explicitly,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}.$$

The determinant of $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is $\det A = ad - bc$. It measures invertibility:

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \qquad \text{provided } ad - bc \neq 0.$$

The identity $\det(AB) = \det(A)\det(B)$ makes determinants especially useful for long products.

Definition A.1. The group $\mathrm{GL}_2(\mathbb{Z})$ consists of the integer matrices with determinant ± 1. Its subgroup $\mathrm{SL}_2(\mathbb{Z})$ consists of those with determinant $+1$.

Proposition A.2 (Integer reversibility)**.** *An integer 2×2 matrix has an inverse with integer entries if and only if its determinant is ± 1.*

Proof. If $\det A = \pm 1$, the displayed inverse formula has integer entries. Conversely, if both A and A^{-1} have integer entries, then $1 = \det(I) = \det(A)\det(A^{-1})$ is a product of two integers. Hence $\det(A) = \pm 1$. □

Big idea

The columns of a matrix in $\mathrm{GL}_2(\mathbb{Z})$ form an integer basis of $\mathbb{Z}^2$: every integer vector has a unique expression as an integer linear combination of those columns. This is the exact algebraic strength supplied by the determinant identity for consecutive convergents.

A.2 Matrices as fractional linear maps

Associate to $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ the Möbius map

$$T_A(t) = \frac{at + b}{ct + d}.$$

A direct substitution shows

$$T_A(T_B(t)) = T_{AB}(t).$$

Thus nested fractional expressions become matrix products. Multiplying A by any nonzero scalar leaves T_A unchanged, but retaining the specific integer matrix remembers determinant and divisibility information that the function alone would forget.

When $ct + d \neq 0$, the quotient rule gives

$$T_A'(t) = \frac{ad - bc}{(ct + d)^2}.$$

Therefore the determinant tells us whether the map is increasing or decreasing. This is the Möbius derivative trick used to locate alternating convergents.

A.3 Continued fractions as iterated products

One partial quotient corresponds to

$$D(a) = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}, \qquad T_{D(a)}(t) = a + \frac{1}{t}.$$

Consequently,

$$T_{D(a_0)\cdots D(a_k)}(t) = [a_0; a_1, \ldots, a_k, t].$$

The convergent recurrences are equivalent to the product identity

$$D(a_0) \cdots D(a_k) = \begin{pmatrix} p_k & p_{k-1} \\ q_k & q_{k-1} \end{pmatrix}.$$

Since every $D(a_i)$ has determinant -1, the product has determinant $(-1)^{k+1}$. Hence

$$p_k q_{k-1} - p_{k-1} q_k = (-1)^{k+1}.$$

This one product packages four facts:

- its first column is the current convergent (p_k, q_k);

- its second column is the preceding convergent (p_{k-1}, q_{k-1});
- its determinant says the two columns form an integer basis;
- its Möbius action inserts an arbitrary remaining tail.

Example A.3. For $[4; 2, 6, 7] = 415/93$,

$$D(4)D(2)D(6)D(7) = \begin{pmatrix} 415 & 58 \\ 93 & 13 \end{pmatrix}.$$

Its inverse is

$$\begin{pmatrix} 13 & -58 \\ -93 & 415 \end{pmatrix}.$$

The top row applied to the first column gives $13(415) - 58(93) = 1$. In the opposite direction, multiplying a vector by the inverse recovers its integer coordinates in the basis of consecutive convergents.

Exercises

1. Multiply three general matrices $D(a_0)D(a_1)D(a_2)$ and identify the two convergents in its columns. [Fluency]
2. Prove $\det(AB) = \det(A)\det(B)$ directly for 2×2 matrices. [Reasoning]
3. Find the inverse of $\left(\begin{smallmatrix} 17 & 7 \\ 12 & 5 \end{smallmatrix}\right)$ and use it to express $(1, 0)$ and $(0, 1)$ as integer combinations of its columns. [Fluency]
4. Verify directly that $T_A \circ T_B = T_{AB}$. Explain why multiplying a matrix by -1 does not change its Möbius map. [Reasoning]
5. Let $P = D(a_1) \cdots D(a_\ell)$. Show that a purely periodic tail $t = [\overline{a_1, \ldots, a_\ell}]$ is a fixed point of T_P, and derive its quadratic equation from the four entries of P. [Reasoning]

Appendix B

Streaming arithmetic with continued fractions

Gosper's homographic and bihomographic algorithms

Suppose a number is arriving one continued-fraction digit at a time. Can we transform it without first converting it to a decimal, and without waiting for the entire infinite input? Bill Gosper's answer is yes. The method reads just enough of the input to prove the next output digit, emits that digit, and then continues.

B.1 The problem—before the machinery

There are two versions.

One input. Given

$$x = [a_0; a_1, a_2, \dots]$$

as a stream, and a linear fractional transformation

$$L(x) = \frac{\alpha x + \beta}{\gamma x + \delta},$$

produce the continued fraction of $L(x)$ from left to right.

Two inputs. Given continued-fraction streams for x and y, produce the continued fraction of a two-variable fractional expression. In particular, $x + y$, $x - y$, xy, and x/y all have the form

$$\Phi(x, y) = \frac{axy + bx + cy + d}{exy + fx + gy + h}.$$

The intended computation is *exact*. It should never guess a digit from a floating-point approximation. Instead, every emitted digit comes with a proof that all inputs compatible with the digits seen so far would produce that same digit.

Big idea

A finite input prefix determines an *interval* of possible values. Map that interval through the desired transformation. If the whole image fits inside one continued-fraction digit cell $[k, k+1)$, the next output digit is certified to be k. If it crosses an integer boundary, read more input and shrink the interval.

The original algorithm appeared as Item 101B of the 1972 MIT memo *HAKMEM*. A modern refinement by Michael J. Collins allows exact intervals themselves to appear in the output stream, resolving boundary cases in which no next digit can be certified after finite time.

B.2 What the unread tail means

The subscript on a tail records the *next* digit, not the first digit of the original number. After the prefix $a_0, \ldots, a_{j-1}$ has been read, write

$$x = [a_0; a_1, \ldots, a_{j-1}, t_j], \qquad t_j = [a_j; a_{j+1}, \ldots]. \tag{B.1}$$

Here t_j is the complete unread remainder of the continued fraction, used as the last entry in the first expression. Before a_j is known, the only universal information is

$$1 \le t_j \le \infty$$

for a positive simple continued fraction. When the next digit $a_j = m$ arrives,

$$t_j = m + \frac{1}{t_{j+1}}, \qquad t_{j+1} = [a_{j+1}; a_{j+2}, \ldots] \in [1, \infty]. \tag{B.2}$$

Thus knowing the single digit m restricts the old tail to $t_j \in (m, m+1]$; absorbing m replaces it by the new unknown tail $t_{j+1} \in [1, \infty]$.

Known prefix versus unknown tail

At the start, $t_0 = x$ and no input digits have been absorbed. After a_0 is read, the state incorporates a_0 and the unknown variable is $t_1 = [a_1; a_2, \ldots]$. After a_1 is read, the state incorporates a_0, a_1 and the unknown variable is t_2, and so on. The state changes; the exact number represented does not.

Let

$$D(m) = \begin{pmatrix} m & 1 \\ 1 & 0 \end{pmatrix}, \qquad M_0 = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}.$$

After j input digits have been absorbed, set

$$M_j = M_0 D(a_0) D(a_1) \cdots D(a_{j-1}) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \tag{B.3}$$

Then the output value still waiting to be written is exactly

$$F_{M_j}(t_j) = \frac{At_j + B}{Ct_j + D}.$$

This is the invariant behind the algorithm.

B.3 The geometry: map an interval, then inspect its floor

For a general certified tail interval $I = [b_0, b_1]$, consider

$$F_M(t) = \frac{At + B}{Ct + D}.$$

Provided $Ct + D$ never vanishes on I, the derivative

$$F'_M(t) = \frac{AD - BC}{(Ct + D)^2} \tag{B.4}$$

has constant sign. Hence F_M is monotone, and the image $F_M(I)$ is simply the interval whose endpoints are $F_M(b_0)$ and $F_M(b_1)$, in the appropriate order. No point between them can escape those endpoint bounds.



If both endpoint images lie in the same floor cell and there is no pole between them, every possible output has floor k.

For the universal interval $[1, \infty]$, the two endpoint values are

$$F_M(1) = \frac{A + B}{C + D}, \qquad F_M(\infty) = \frac{A}{C}.$$

An endpoint at infinity may be open, and a finite continued fraction may end; exact implementations retain this endpoint information. The essential test is unchanged:

$$F_M(I) \subseteq [k, k+1) quad \Longrightarrow \quad \text{the next output digit is } k.$$

If the image crosses an integer boundary, or if a pole is possible, no digit is yet justified. We must refine the input interval.

B.4 The two algebraic moves

Absorb the next input digit. If $a_j = m$, substitute $t_j = m + 1/t_{j+1}$ into the current map. Clearing the factor t_{j+1} gives

$$F_M\left(m + \frac{1}{t_{j+1}}\right) = \frac{(Am + B)t_{j+1} + A}{(Cm + D)t_{j+1} + C}.$$

Thus the exact state update is

$$\boxed{\mathcal{I}_m(M) = MD(m) = \begin{pmatrix} Am + B & A \\ Cm + D & C \end{pmatrix}.} \tag{B.5}$$

Emit a certified output digit. If the image interval lies in $[k, k+1)$, write the digit k. The unread part of the output continued fraction is then $1/(F_M - k)$, represented by

$$\boxed{\mathcal{E}_k(M) = \begin{pmatrix} 0 & 1 \\ 1 & -k \end{pmatrix} M = \begin{pmatrix} C & D \\ A - kC & B - kD \end{pmatrix}.} \tag{B.6}$$

The homographic loop

Maintain the statement: "the unread output is $F_M(t_j)$ for every $t_j \in I$."

1. Map the current certified interval I through F_M by evaluating its endpoints.
2. If the image lies in one floor cell $[k, k+1)$, emit k, apply $M \leftarrow \mathcal{E}_k(M)$, and replace the output interval by $1/(F_M(I) - k)$.
3. Otherwise read $a_j = m$, use $t_j = m + 1/t_{j+1}$, apply $M \leftarrow \mathcal{I}_m(M)$, and return to the universal interval $t_{j+1} \in [1, \infty]$.
4. Repeat. A common nonzero factor may be removed from the four matrix entries without changing the map.

B.5 Worked stream: the first digits of $\pi/2$

Here $L(x) = x/2$, so $M_0 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$, and the input is

$$\pi = [3; 7, 15, 1, 292, \dots].$$

After reading $a_0 = 3$ and $a_1 = 7$, the unknown variable is $t_2 = [15; 1, 292, \dots] \in [1, \infty]$, and

$$M_2 = M_0 D(3) D(7) = \begin{pmatrix} 22 & 3 \\ 14 & 2 \end{pmatrix}.$$

The first row below maps the endpoint tails 1 and ∞. Each following row applies (B.6); no further input digit is read.

event	current matrix M	image of the possible tail interval
read $3, 7$	$\left(\begin{smallmatrix} 22 & 3 \\ 14 & 2 \end{smallmatrix}\right)$	$[25/16,\ 11/7)$: emit 1
emit 1	$\left(\begin{smallmatrix} 14 & 2 \\ 8 & 1 \end{smallmatrix}\right)$	$(7/4,\ 16/9]$: emit 1
emit 1	$\left(\begin{smallmatrix} 8 & 1 \\ 6 & 1 \end{smallmatrix}\right)$	$[9/7,\ 4/3)$: emit 1
emit 1	$\left(\begin{smallmatrix} 6 & 1 \\ 2 & 0 \end{smallmatrix}\right)$	$(3,\ 7/2]$: emit 3

After these four emissions the state is $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right)$: the unread output is $2t_2$. Only now do we read the next input digits, 15 and 1; absorbing them maps the new unknown tail into $[31, 32)$ and certifies 31. Therefore

$$\frac{\pi}{2} = [1; 1, 1, 3, 31, \dots].$$

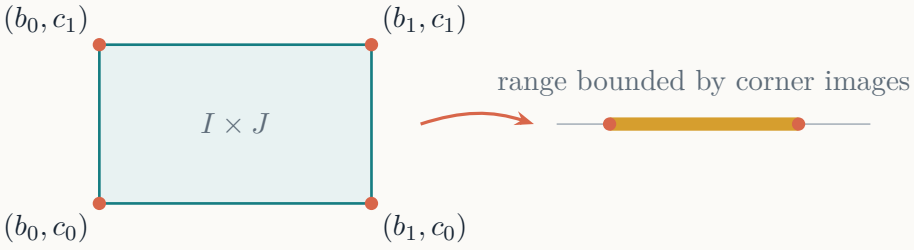
The calculation has used exact rational endpoints throughout. The decimal digits of π never entered.

B.6 Two inputs: rectangles replace intervals

Now suppose prefixes of two inputs have been read. Their unread tails, say u_j and v_ℓ, range independently over certified intervals I and J. The unread output has the form

$$B_{\mathcal{B}}(u,v) = \frac{auv + bu + cv + d}{euv + fu + gv + h}, \qquad \mathcal{B} = (a,b,c,d;\, e,f,g,h). \tag{B.7}$$

Thus the possible input is a rectangle $I \times J$, not a line segment.



Assume the denominator has no zero on the rectangle. For a fixed v, the map is linear fractional in u, so its extrema along that horizontal segment occur at the two ends. On either vertical edge it is linear fractional in v, so the remaining extrema occur at the edge endpoints. Consequently the entire range is bounded by the images of the four corners. In particular, for $I = J = [1, \infty]$ these values are

$$\frac{a}{e}, \qquad \frac{a+b}{e+f}, \qquad \frac{a+c}{e+g}, \qquad \frac{a+b+c+d}{e+f+g+h}, \tag{B.8}$$

with the usual interpretation of infinite and open endpoints.

The four basic operations differ only in their starting coefficients:

desired output	$(a,b,c,d;\, e,f,g,h)$
$x+y$	$(0,1,1,0;\, 0,0,0,1)$
$x-y$	$(0,1,-1,0;\, 0,0,0,1)$
xy	$(1,0,0,0;\, 0,0,0,1)$
x/y	$(0,1,0,0;\, 0,0,1,0)$

If the next u-digit is m, substitute $u = m + 1/u'$; if the next v-digit is n, substitute $v = n + 1/v'$. Collecting coefficients gives

$$\mathcal{I}^u_m(\mathcal{B}) = (ma + c,\ mb + d,\ a,\ b;\ me + g,\ mf + h,\ e,\ f), \tag{B.9}$$

$$\mathcal{I}^v_n(\mathcal{B}) = (na + b,\ a,\ nc + d,\ c;\ ne + f,\ e,\ ng + h,\ g). \tag{B.10}$$

When the four corner images all lie in one floor cell $[k, k+1)$, emit k and invert the fractional part:

$$\mathcal{E}_k(\mathcal{B}) = (e,f,g,h;\ a - ke, b - kf, c - kg, d - kh). \tag{B.11}$$

The geometry and the logic are exactly as before. If the image of the current rectangle crosses an integer boundary, absorb from one input and replace the rectangle by a thinner one; alternate inputs, or choose the one expected to shrink the image more. If the denominator may vanish inside the rectangle, no output is safe until further input or separate algebra resolves the issue.

Big idea

The four coefficients in the one-input problem, or eight in the two-input problem, are bookkeeping for a moving interval image. Matrix formulas make the updates fast, but the proof of correctness is geometric: every possible tail lies in the current interval or rectangle, and every emitted digit is shared by its entire image.

B.7 The boundary problem and the interval repair

The floor function jumps at every integer. This creates a subtle failure mode. Feed two *independent* copies of the continued fraction of $\sqrt{2}$ to the multiplication algorithm. The actual product is 2, but every finite pair of independent enclosures allows one copy to move a little below $\sqrt{2}$ and the other a little above it. The resulting product rectangle can straddle 2. Its image then meets both floor cells $[1, 2)$ and $[2, 3)$, so the rule cannot certify even the first digit.

This is an information problem, not numerical instability. The two stream variables have forgotten that they denote the same algebraic number. Retaining the relation $x = y$ reduces the expression to x^2, and the separate fact $x^2 = 2$ resolves the equality immediately.

Collins's interval-valued refinement distinguishes a continued fraction from its computational stream. A stream item may be either

- an explicit partial quotient, or
- an exact rational interval bounding the current unread value.

If no digit is forced, the algorithm may output a tighter certified interval instead of waiting silently. A consumer asks for an error tolerance ε and reads until the resulting value interval has width below ε. Exact endpoints and their open-or-closed status preserve the no-roundoff character of Gosper's method while guaranteeing useful progress at an integer boundary.

Digits and numerical accuracy are different requests

There may be no finite proof of the next continued-fraction digit even when a very accurate numerical enclosure is easy. A robust implementation therefore asks for a certified error tolerance, not an unconditional number of output digits. Away from boundaries, intervals quickly collapse into ordinary partial quotients and recover Gosper's original stream.

B.8 Exercises

Exercises

1. Suppose only the prefix $[2; 3]$ of x is known. Write $x = [2; 3, t_2]$ with $t_2 \in [1, \infty]$, evaluate the two endpoint values, and prove that $x \in [9/4, 7/3)$. [Fluency]
2. Prove the endpoint-image rule using (B.4). Apply it to $F(t) = (22t + 3)/(14t + 2)$ on $[1, \infty]$ and certify its floor. [Reasoning]
3. Derive the absorption and emission updates (B.5) and (B.6) directly by substitution.

Express each as matrix multiplication. [Fluency]

4. Continue the $\pi/2$ calculation: from $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, absorb 15 and 1 and show that the new image lies in $[31, 32)$. [Reasoning]

5. Prove the four-corner range rule for a bihomographic map on a finite rectangle with no denominator zero. Why is checking four corners not enough if a pole lies inside? [Reasoning]

6. Derive (B.9) and (B.10). Verify that the four initial states really represent $+$, $-$, $\times$, and $/$. [Fluency]

7. Explain why independent intervals for two copies of $\sqrt{2}$ can straddle the boundary 2 forever when computing $\sqrt{2}\sqrt{2}$. State the extra algebraic information that resolves the equality. [Reasoning]

8. Design a lazy implementation whose output type is either `Digit(k)` or `Bound(lo,hi)`. State invariants that make every emitted item a certificate, then test all four operations on rational and quadratic inputs. [Project]

Appendix C

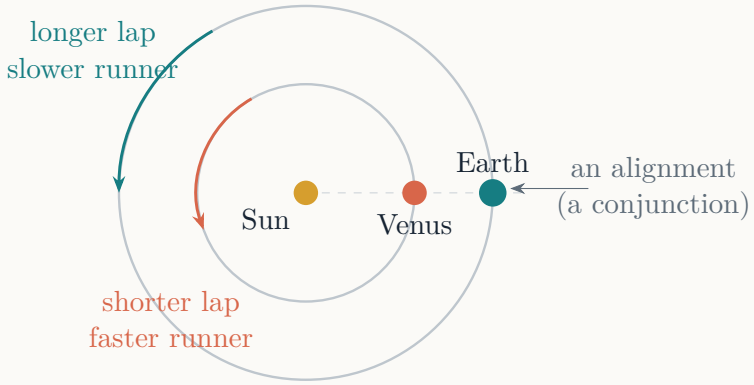
Venus writes Euclid in the sky

Three clocks, five alignments, and the Dresden Codex

Continued fractions are useful whenever two repeating motions almost return to their starting arrangement at the same time. Venus gives a particularly visible example. Small Fibonacci numbers describe a near-repeat of the modern planetary motions, while related integers organize an exact calendar table in the Dresden Codex. We will build the astronomical vocabulary from scratch and keep the approximate observation separate from the exact calendar arithmetic.

C.1 Begin with two runners on circular tracks

Imagine Earth and Venus as two runners circling the Sun. Venus uses the inner track and completes a lap sooner, so it repeatedly catches and passes Earth. At the instant shown below, the Sun, Venus, and Earth lie on the same ray.



Here are the three time intervals we need.

Symbol	Everyday meaning	Astronomical name and approximate value
E	time for Earth to make one Sun-centered lap	Earth's *sidereal period*: 365.256 days
V	time for Venus to make one Sun-centered lap	Venus's *sidereal period*: 224.7 days
S	time from one Earth–Venus alignment to the next	Venus's *synodic period*: 583.92 days

"Sidereal" means that a full lap is measured relative to the distant stars. A "conjunction" is an alignment; in the alignment drawn above, Venus lies between Earth and the Sun, so astronomers call it an *inferior conjunction*. "Synodic period" simply means the waiting time until the same kind of alignment occurs again.[1]

C.2 The overtaking clock

Earth completes $1/E$ laps per day, while Venus completes $1/V$. Because Venus is faster, it gains on Earth at the rate

$$\frac{1}{V} - \frac{1}{E} \quad \text{laps per day.}$$

The next alignment occurs when Venus has gained exactly one whole lap. Hence

$$\frac{1}{S} = \frac{1}{V} - \frac{1}{E}. \tag{C.1}$$

This is the same "lapping time" calculation used for two runners on a track.

Put $r = E/V$: it compares the length of an Earth lap in time with a Venus lap. Multiplying (C.1) by E gives

$$\frac{S}{E} = \frac{1}{r-1} =: T(r). \tag{C.2}$$

The measured ratio $E/V \approx 1.6255$ has the simple, very accurate fraction $13/8$ nearby. If we replace the measured ratio temporarily by this ideal one, then

$$T\left(\frac{13}{8}\right) = \frac{8}{5}.$$

In plain language:

- during about eight Earth laps, Venus makes about thirteen laps; and
- in that time Venus gains $13 - 8 = 5$ laps on Earth, producing about five repeat alignments.

None of these counts is exact for the physical planets. Their usefulness is that $13/8$ is an excellent approximation with a tiny denominator.

[1] For the modern mean values, see NASA, *2012 Transit of Venus* and NASA SP–461, *Pioneer Venus*. The real orbits are neither perfectly circular nor perfectly uniform; rounded mean periods are all we need here.

Why the golden ratio appears

The positive fixed point of the overtaking map $T(r) = 1/(r-1)$ satisfies

$$r = \frac{1}{r-1}, \qquad r^2 - r - 1 = 0,$$

so it is $\varphi = (1+\sqrt{5})/2$. More generally,

$$T\left(\frac{F_{n+1}}{F_n}\right) = \frac{F_n}{F_{n-1}}.$$

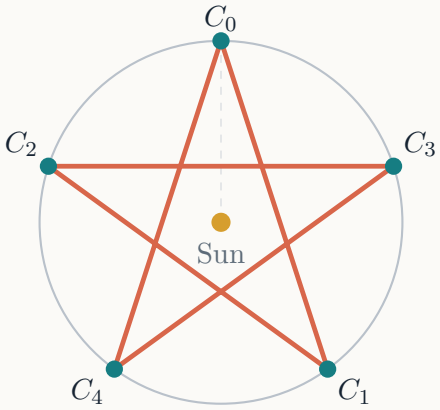
The calculation of one lapping time walks backward through consecutive Fibonacci ratios.

C.3 Why five alignments draw a star

During one alignment-to-alignment interval, Earth travels about $S/E \approx 8/5$ laps. One complete lap returns to the same direction, so only the leftover $3/5$ of a lap changes the direction of the alignment ray. The five successive directions are therefore approximately

$$0, \quad \frac{3}{5}, \quad \frac{1}{5}, \quad \frac{4}{5}, \quad \frac{2}{5} \pmod 1.$$

Connecting those directions in time order draws a five-pointed star.



Each C_j marks the direction of an alignment, not the path traveled by Venus. The actual fifth return misses the starting direction slightly.

The geometry of a perfectly regular pentagram contains the golden ratio exactly. The planetary star is only approximate: because 13/8 is not the exact period ratio, the alignment points drift and the whole pattern slowly turns over repeated eight-year cycles.[2]

[2]The European Southern Observatory's educational note *The Pentagram of Venus* illustrates the five alignments and the small miss after eight years.

C.4 The Euclidean algorithm inside $13 : 8$

The approximation contains the short Euclidean chain

$$\begin{aligned} 13 &= 1 \cdot 8 + 5, \quad 8 = 1 \cdot 5 + 3, \\ 5 &= 1 \cdot 3 + 2, \quad 3 = 1 \cdot 2 + 1, \\ 2 &= 2 \cdot 1. \end{aligned}$$

Consequently

$$\frac{13}{8} = [1; 1, 1, 1, 2] = [1; 1, 1, 1, 1, 1], \qquad \frac{8}{5} = [1; 1, 1, 2] = [1; 1, 1, 1, 1].$$

These are consecutive convergents of $\varphi = [1; \overline{1}]$. The orbit ratio 13/8, the difference 5, the step 3/5, and the pentagram are not independent coincidences: they are different consequences of one good rational approximation.

C.5 The Dresden Codex: an exact calendar model

The modern calculation above starts with measured periods and obtains approximations. The Maya Venus Table in the Dresden Codex instead uses chosen whole-day counts for calendrical bookkeeping. In particular:

- 584 days is the table's rounded alignment-to-alignment Venus cycle;
- 365 days is one *Haab*, the Maya solar-year calendar; and
- 260 days is one Tzolk'in, a repeating named-day cycle.

Five pages of the table present one 584-day Venus cycle apiece. The five cycles make an eight-Haab block because

$$5 \cdot 584 = 8 \cdot 365 = 2920. \tag{C.3}$$

This equality is exact arithmetic inside the calendar model, not an assertion that the physical Venus cycle is exactly 584 days.[3]

To make the 2920-day block agree again with the 260-day named-day cycle, take a least common multiple. Since $\gcd(2920, 260) = 20$,

$$\operatorname{lcm}(2920, 260) = \frac{2920 \cdot 260}{20} = 37960.$$

At that common return,

$$65 \cdot 584 = 104 \cdot 365 = 146 \cdot 260 = 13 \cdot 2920 = 37960. \tag{C.4}$$

[3] See the Saxon State and University Library Dresden's description of the Dresden Maya Codex, especially the Venus pages 24 and 46–50. For the page structure and cumulative day counts, see Akira Kaneko, "Identification of Postclassic Maya Constellations from the Venus Pages of the Dresden Codex," *Estudios de Cultura Maya* **35** (2010).

The same numbers are doing different jobs

In the modern orbital approximation, 13 counts Venus's Sun-centered laps in about eight Earth laps. In (C.4), 13 counts complete 2920-day calendar blocks in a 37960-day table. The identity $13 = 8 + 5$ is a delightful number-theoretic lens, but the manuscript alone does not show that Maya astronomers formulated their table using Fibonacci numbers or continued fractions.

Finally, rounding has a cost. The modern mean alignment cycle is about 583.92 days, so the calendar value 584 gains roughly

$$65(584 - 583.92) \approx 5.2 \text{ days}$$

over 65 cycles. Scholars study correction schemes in the table. Our exact identity (C.4) describes its integer framework, not a perfectly uniform planet.

Exercises

1. In your own words, distinguish a planet's Sun-centered lap time from the waiting time between two Earth–planet alignments. Which are E, V, and S? [Fluency]
2. Derive (C.1) by comparing lap rates. Then derive (C.2) and verify that $T(13/8) = 8/5$. [Reasoning]
3. Starting from a step of $3/5$ of a turn, list the first five positions modulo 1. Why does joining them chronologically make a star rather than the boundary of a pentagon? [Reasoning]
4. Run the Euclidean algorithm on $(13, 8)$. Read its quotients as the continued fraction of $13/8$, and explain why replacing the final 2 by $1, 1$ gives the same rational number. [Reasoning]
5. Factor 2920 and 260, compute their gcd and lcm, and recover every multiplier in (C.4). [Fluency]
6. Prove $T(F_{n+1}/F_n) = F_n/F_{n-1}$. Show that the positive fixed point of T is φ, and connect this with $[1; \overline{1}]$. [Reasoning]
7. Using 583.92 days as the modern mean alignment cycle, compare the elapsed time for 5, 65, and 650 cycles with the nominal 584-day count. Which error is negligible for a sketch but serious for a long calendar? [Exploration]

Appendix D

Selected hints and solutions

For checking arguments, not replacing them

Only selected exercises are treated here. A useful routine is: write a complete attempt, mark the exact step where you are stuck, then consult only the first relevant sentence below.

Chapter 1

1. Euclid gives

$$2024 = 2(748) + 528, \quad 748 = 528 + 220, \quad 528 = 2(220) + 88,$$

$$220 = 2(88) + 44, \quad 88 = 2(44),$$

so the gcd is 44.

2. The divisions are $527 = 341 + 186$, $341 = 186 + 155$, $186 = 155 + 31$, $155 = 5 \cdot 31$. Reverse:

$$31 = 186 - 155 = 2(186) - 341 = 2(527 - 341) - 341 = 2 \cdot 527 - 3 \cdot 341.$$

4. One solution is $35(-5) + 22(8) = 1$. All solutions are

$$x = -5 + 22t, \qquad y = 8 - 35t \qquad (t \in \mathbb{Z}).$$

To prove completeness, subtract two solutions and use $35(x - x') = -22(y - y')$ with $\gcd(35, 22) = 1$.

Chapter 2

3. Write $c = ar = bs$. Since $a \mid bs$ and $\gcd(a, b) = 1$, repeated Euclid's lemma (or Bézout) gives $a \mid s$. Thus $s = at$ and $c = abt$.

4. If $\sqrt{p} = a/b$ in lowest terms, then $a^2 = pb^2$. The exponent of p on the left is even; on the right it is odd. Unique factorization forbids this.

6. The primes at most 100 are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.$$

Every composite at most 100 has a prime factor at most $\sqrt{100} = 10$, so sieving by the primes $2, 3, 5, 7$ crosses out every composite.

7. No: $5! + 1 = 121 = 11^2$. Euclid's number only needs a prime divisor outside the proposed list; it need not itself be prime.

9. We already know that $a = \sqrt{2}$ is irrational. By the definition of the logarithm, $a^b = 3$. If $b = p/q$ were rational with $q > 0$, then $(\sqrt{2})^{p/q} = 3$; raising both sides to $2q$ would give $2^p = 3^{2q}$, contrary to unique factorization. Hence b is irrational. Unlike the proof by cases, this argument names a particular pair and proves that it works; it does not leave the witness dependent on an undecided alternative.

Chapter 3

1. Euclid gives

$$355/113 = [3; 7, 16],$$

with convergents 3, 22/7, and 355/113.

2. The convergents are

$$1,\ 2,\ \frac{3}{2},\ \frac{5}{3},\ \frac{8}{5},\ \frac{13}{8}.$$

Numerators and denominators are consecutive Fibonacci numbers; the recurrence with every $a_k = 1$ proves it.

4. Since $a_n = (a_n - 1) + 1/1$ at the bottom level,

$$[a_0; \dots, a_n] = [a_0; \dots, a_n - 1, 1].$$

Requiring the last partial quotient to exceed 1 selects a unique finite expansion.

6. With $\alpha \approx 0.2421988$, the continued fraction begins

$$\alpha = [0; 4, 7, 1, 3, 5, \dots],$$

so its first four nonzero convergents are

$$\frac{1}{4},\qquad \frac{7}{29},\qquad \frac{8}{33},\qquad \frac{31}{128}.$$

The Julian calendar uses $1/4$ leap day per year. A century-based cycle has $q = 100Q$, so we instead approximate

$$100\alpha \approx 24.21988 = [24; 4, 1, 1, 4, \dots].$$

Its first nonintegral convergent is $97/4$; dividing the denominator scale by 100 gives $97/400$, the Gregorian rate. The corresponding mean years are

$$365 + \frac{1}{4} = 365.25,\qquad 365 + \frac{97}{400} = 365.2425,\qquad 365 + \frac{31}{128} = 365.2421875.$$

Relative to the fixed approximation 365.2421988, their annual errors in days are about 7.80×10^{-3}, 3.01×10^{-4}, and -1.13×10^{-5}, respectively. Thus $31/128$ is numerically better, whereas $97/400$ fits a convenient rule organized by fours and centuries.

8. Euclid gives $527/341 = [1;1,1,5]$, whose convergents are $1, 2, 3/2, 17/11$. The penultimate convergent is $p_2/q_2 = 3/2$, so the remainder formula gives

$$\gcd(527, 341) = r_2 = 2 \cdot 527 - 3 \cdot 341 = 31.$$

9. The first six convergents of $\sqrt{3} = [1;\overline{1,2}]$ are

$$1, \quad 2, \quad \frac{5}{3}, \quad \frac{7}{4}, \quad \frac{19}{11}, \quad \frac{26}{15}.$$

They satisfy

$$1 < \frac{5}{3} < \frac{19}{11} < \sqrt{3} < \frac{26}{15} < \frac{7}{4} < 2.$$

For even k, $M_k'(t) < 0$, so the finite-tail value lies above $M_k(\infty) = c_k$; for odd k, the derivative is positive and the inequality reverses. Applying this once to x and once to c_{k+2} gives the two chains.

10. For $k = 0$ the formula is just $D(a_0) = \left(\begin{smallmatrix} p_0 & p_{-1} \\ q_0 & q_{-1} \end{smallmatrix}\right)$. If it holds at k, right multiplication by $D(a_{k+1})$ replaces the first column by $a_{k+1}(p_k, q_k)^T + (p_{k-1}, q_{k-1})^T$ and moves the old first column into second place. These are exactly the recurrences. The determinant is $(-1)^{k+1}$, so the inverse formula introduces no denominators.

11. For $1/7$, the remainders are $1, 3, 2, 6, 4, 5, 1, \ldots$, producing $0.\overline{142857}$. For $5/12$, they are $5, 2, 8, 8, \ldots$, producing $0.41\overline{6}$. The first repeated remainder marks the start and length of the repeating block.

15. Since

$$D(1)D(2) = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix},$$

the repeating tail obeys $t = (3t + 1)/(2t + 1)$. Hence $2t^2 - 2t - 1 = 0$, and positivity selects $t = (1 + \sqrt{3})/2$.

16. For $x = 1 + \sqrt{2}$, the conjugate $x' = 1 - \sqrt{2}$ lies in $(-1, 0)$; for $x = (1 + \sqrt{5})/2$, the conjugate is $(1 - \sqrt{5})/2$, again in $(-1, 0)$. Both numbers exceed 1, so Galois's criterion applies. Indeed their expansions are $[\overline{2}]$ and $[\overline{1}]$, respectively.

17. Factoring the difference of squares gives

$$\left|\sqrt{D} - p/q\right| = \frac{|D - p^2/q^2|}{\sqrt{D} + p/q} = \frac{1}{q^2(\sqrt{D} + p/q)}.$$

The denominator factor $\sqrt{D} + p/q$ exceeds 2, so Legendre's criterion forces p/q to be a convergent of $\sqrt{D}$.

18. We have $\sqrt{3} = [1;\overline{1,2}]$, of even period length 2; one period therefore gives the $+1$ solution $2^2 - 3(1)^2 = 1$. The first columns of the first three powers of $\left(\begin{smallmatrix} 2 & 3 \\ 1 & 2 \end{smallmatrix}\right)$ are

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad \begin{pmatrix} 7 \\ 4 \end{pmatrix}, \qquad \begin{pmatrix} 26 \\ 15 \end{pmatrix},$$

and each satisfies $p^2 - 3q^2 = 1$.

19. A finite computation verifies only the displayed finite averages; an asymptotic conclusion requires control of every later partial quotient. Khinchin's theorem excludes only a measure-zero

exceptional set and supplies no known test proving that a named constant such as π lies outside it. For e, the repeating pattern of positions gives

$$a_{3j-2} = 1, \qquad a_{3j-1} = 2j, \qquad a_{3j} = 1,$$

so $a_1 \cdots a_{3m} = 2^m m!$. Taking the $(3m)$th root and applying $m! \sim \sqrt{2\pi m}(m/e)^m$ yields

$$G_{3m}(e) = (2^m m!)^{1/(3m)} \sim (2m/e)^{1/3} \to \infty.$$

The denominator recurrence gives $q_{3m} \geq a_1 \cdots a_{3m} = 2^m m!$, so $q_{3m}^{2/(3m)} \to \infty$ too. Thus e is provably on the exceptional side of both laws.

Chapter 4

1. Extended Euclid yields

$$1 = 11 \cdot 101 - 30 \cdot 37,$$

so $37^{-1} \equiv -30 \equiv 71 \pmod{101}$.

2. Divide $18x \equiv 30 \pmod{42}$ by 6 while reducing the modulus: $3x \equiv 5 \pmod{7}$. Since $3^{-1} \equiv 5$, $x \equiv 4 \pmod{7}$. Modulo 42 the six solutions are $4, 11, 18, 25, 32, 39$.

5. Here $N = 504$. One convenient construction is

$$x = 2(72)(4) + 3(63)(7) + 4(56)(5) = 3019 \equiv 499 \pmod{504}.$$

The multipliers $4, 7, 5$ invert $72, 63, 56$ modulo $7, 8, 9$, respectively.

Chapter 5

1. The binary length-3 words form four orbits:

$$\{000\}, \quad \{111\}, \quad \{001, 010, 100\}, \quad \{011, 110, 101\}.$$

3. If $ik \equiv jk \pmod{p}$, then $p \mid k(i-j)$. As $p \nmid k$, Euclid's lemma gives $p \mid i - j$. For $0 \leq i, j < p$, this forces $i = j$.

5. Orbit sizes divide 6, and binary examples realize $1, 2, 3, 6$: 000000, 010101, 001001, and 000001.

Chapter 6

1. $(\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\}$. Every nonidentity element squares to 1, so their orders are 2; the identity has order 1.

4. If $aH \cap bH$ contains x, write $x = ah_1 = bh_2$. Then $a = bh_2h_1^{-1}$, so every $ah \in aH$ equals $b(h_2h_1^{-1}h) \in bH$. Thus $aH \subseteq bH$; symmetry gives equality.

5. For $g \neq e$, the order of g divides the prime $|G|$, so it is $|G|$. Hence $G = \langle g \rangle$.

Chapter 7

2. Legendre gives

$$v_2(100!) = 50 + 25 + 12 + 6 + 3 + 1 = 97,$$
$$v_5(100!) = 20 + 4 = 24.$$

Each trailing zero consumes one 2 and one 5, so there are 24.

4. Choose n objects from two disjoint groups of n. If k come from the first group, then $n-k$ come from the second. Summing over k gives

$$\sum_k \binom{n}{k}\binom{n}{n-k} = \binom{2n}{n}.$$

7. Legendre gives

$$v_p\binom{2n}{n} = \sum_{j\geq 1}\left(\left\lfloor \frac{2n}{p^j}\right\rfloor - 2\left\lfloor \frac{n}{p^j}\right\rfloor\right).$$

Each summand is 0 or 1, and there are at most $\left\lfloor \log_p(2n)\right\rfloor$ nonzero candidates. This is the exponent of p in the lcm, so the valuation of the binomial coefficient is no larger for every prime p.

Chapter 8

2. The sum of the $2n+1$ coefficients in row $2n$ is 4^n. The central coefficient is largest, so it is at least the average $4^n/(2n+1)$ and strictly below the whole sum.

3. For $n < p \leq 2n$, the numerator $(n+1)\cdots(2n)$ contains exactly one multiple of p, while $n!$ contains none. Because $p^2 > 2n$ for $p > n \geq 1$, the valuation is exactly one.

7 (hint). Insert $x = p_n$ into both Chebyshev bounds to compare n with $p_n/\log p_n$. First obtain coarse polynomial bounds on p_n so that $\log p_n$ can be compared with $\log n$, then substitute back.

Appendix B: continued-fraction arithmetic

1. With $t_2 \in [1,\infty]$,

$$x = 2 + \frac{1}{3 + 1/t_2}.$$

This expression increases with t_2. At $t_2 = 1$ it is $9/4$; as $t_2 \to \infty$ it approaches $7/3$. For an infinite continued fraction the second endpoint is not attained, so $x \in [9/4, 7/3)$.

2. The denominator is positive for $t \geq 1$, and

$$F'(t) = \frac{22\cdot 2 - 3\cdot 14}{(14t+2)^2} = \frac{2}{(14t+2)^2} > 0.$$

Thus the range is determined by the endpoint values $F(1) = 25/16$ and $F(\infty) = 11/7$. The whole interval lies in $[1,2)$, so the certified output digit is 1.

3. Substituting $t = m + 1/u$ into $F_M(t) = (At + B)/(Ct + D)$ and clearing the factor u gives

$$\frac{(Am + B)u + A}{(Cm + D)u + C},$$

which is represented by $MD(m)$. After emitting k, the unread output is $1/(F_M - k)$, represented by $\left(\begin{smallmatrix} 0 & 1 \\ 1 & -k \end{smallmatrix}\right)M$.

4. From $M = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, absorption gives

$$MD(15) = \begin{pmatrix} 30 & 2 \\ 1 & 0 \end{pmatrix}, \qquad MD(15)D(1) = \begin{pmatrix} 32 & 30 \\ 1 & 1 \end{pmatrix}.$$

The latter map has range $[31, 32)$ on $[1, \infty)$, so the next output digit is 31.

5. For each fixed v, the map in u is linear fractional and hence monotone unless it is constant; its extrema on a horizontal slice occur on the two vertical edges. On either edge, the remaining function of v is again linear fractional, so its extrema occur at the two endpoints. Therefore all global extrema are corner values. If the denominator vanishes inside the rectangle, the map is not continuous there and can become unbounded, so corner values alone give no range certificate.

7. Treating the two copies as independent discards the relation $x = y$. Every finite continued-fraction prefix gives a nontrivial interval containing $\sqrt{2}$, so the product box permits values slightly below and slightly above 2. Since the floor jumps at 2, it is never certified from such boxes. Retaining the dependency $x = y$ changes the expression to x^2; using the known equation $x^2 = 2$ resolves it immediately. Without that extra algebraic relation, an interval-valued output can still certify arbitrarily small numerical error.

Appendix C: Venus and astronomical cycles

1. E is Earth's Sun-centered lap time and V is Venus's Sun-centered lap time. The quantity S measures something different: the waiting time between successive Earth–Venus alignments.

2. Venus and Earth move through $1/V$ and $1/E$ laps per day. Their relative phase therefore increases at $1/V - 1/E$ revolutions per day, so one relative revolution takes S days with $1/S = 1/V - 1/E$. If $r = E/V$, then

$$\frac{E}{S} = \frac{E}{V} - 1 = r - 1, \qquad \frac{S}{E} = \frac{1}{r - 1}.$$

Substitution gives $T(13/8) = 1/(13/8 - 1) = 8/5$.

3. Repeated addition of $3/5$ modulo 1 gives

$$0,\ \frac{3}{5},\ \frac{6}{5} \equiv \frac{1}{5},\ \frac{4}{5},\ \frac{7}{5} \equiv \frac{2}{5},$$

then returns to 0. Each step skips one neighboring vertex of the regular pentagon, so the chronological edges are the diagonals and form a pentagram.

5. The factorizations are

$$2920 = 2^3 \cdot 5 \cdot 73, \qquad 260 = 2^2 \cdot 5 \cdot 13.$$

Thus the gcd is 20 and the lcm is $2^3 \cdot 5 \cdot 13 \cdot 73 = 37960$. Division by $584, 365, 260$, and 2920 gives the multipliers $65, 104, 146$, and 13.

6. The Fibonacci recurrence gives

$$T\left(\frac{F_{n+1}}{F_n}\right) = \frac{1}{F_{n+1}/F_n - 1} = \frac{F_n}{F_{n+1} - F_n} = \frac{F_n}{F_{n-1}}.$$

The positive solution of $r = 1/(r-1)$ is $r = (1+\sqrt{5})/2 = \varphi$. Its identity $\varphi = 1 + 1/\varphi$ is exactly the infinite continued fraction $[1; \overline{1}]$.

7. The nominal count gains 0.08 day per cycle, hence 0.4 day after 5 cycles, 5.2 days after 65, and 52 days after 650.

Appendix E

Reference sheets

Definitions, algorithms, and proof moves

Core notation

Notation	Meaning
$a \mid b$	$b = ak$ for some integer k
$\gcd(a, b)$	greatest positive common divisor
$a \equiv b \pmod{n}$	$n \mid (a - b)$
$(\mathbb{Z}/n\mathbb{Z})^\times$	invertible residue classes modulo n
$\varphi(n)$	number of units modulo n
$v_p(m)$	exponent of p in m
$\pi(x)$	number of primes at most x
$\vartheta(x)$	$\sum_{p \le x} \log p$
$\psi(x)$	$\sum_{p^k \le x} \log p$
$f(x) \asymp g(x)$	their ratio stays between positive constants eventually

Algorithms at a glance

Extended Euclid. Divide until zero; the last nonzero remainder is the gcd; substitute backward to obtain $ax + by = \gcd(a, b)$.

Sieve of Eratosthenes. To find the primes through N, repeatedly circle the least uncrossed integer and cross out its larger multiples. Stop after the sieving primes $p \le \sqrt{N}$; a composite at most N must have a prime factor in that range.

Existence versus a witness. The classical dichotomy for $x = (\sqrt{2})^{\sqrt{2}}$ proves that one of $(\sqrt{2}, \sqrt{2})$ and $(x, \sqrt{2})$ has a rational power, but does not say which. The explicit choice $a = \sqrt{2}$, $b = \log_{\sqrt{2}} 3$ gives $a^b = 3$, and unique factorization proves that b is irrational.

Continued fraction. Repeatedly take the integer part, subtract it, and invert. Convergents satisfy

$$p_k = a_k p_{k-1} + p_{k-2}, \qquad q_k = a_k q_{k-1} + q_{k-2}.$$

The recurrences are the columns of the matrix product

$$\begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_k & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} p_k & p_{k-1} \\ q_k & q_{k-1} \end{pmatrix},$$

whose determinant is $(-1)^{k+1}$. Each factor lies in $\mathrm{GL}_2(\mathbb{Z})$; a product with an even number of factors lies in $\mathrm{SL}_2(\mathbb{Z})$. For $A/B = [a_0; \ldots, a_n]$, the Euclidean remainders obey

$$r_k = (-1)^k (q_k A - p_k B).$$

For irrational x, the zero-indexed convergents form alternating monotone bounds:

$$c_0 < c_2 < c_4 < \cdots < x < \cdots < c_5 < c_3 < c_1.$$

Legendre's test says that a reduced fraction satisfying $|x - p/q| < 1/(2q^2)$ must be a convergent. Lagrange's theorem says that a simple continued fraction is eventually periodic exactly for a quadratic irrational; Galois says it is purely periodic exactly when $x > 1$ and $-1 < x' < 0$.

Typical information rate. For almost every x, the Khinchin–Lévy law gives

$$\lim_{n\to\infty} q_n^{1/n} = e^{\pi^2/(12\log 2)} \approx 3.275823, \qquad \lim_{n\to\infty} |x - p_n/q_n|^{-1/n} = e^{\pi^2/(6\log 2)} \approx 10.731.$$

So a continued-fraction term earns about 1.0306 decimal digits on average. For the first 100,000 partial quotients of π, computation gives $G_n(\pi) \approx 2.67956$ and $q_n^{2/n} \approx 10.67835$; neither limiting law has been proved for π. In contrast, $e = [2; 1, 2, 1, 1, 4, 1, 1, 6, \ldots]$ has $G_{3m}(e) \sim (2m/e)^{1/3} \to \infty$ and $q_{3m}^{2/(3m)} \to \infty$.

Gosper's streaming arithmetic. Represent a unary rational transformation by $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$. Reading an input digit m and emitting a certified output digit k update the state by

$$M \longmapsto M \begin{pmatrix} m & 1 \\ 1 & 0 \end{pmatrix}, \qquad M \longmapsto \begin{pmatrix} 0 & 1 \\ 1 & -k \end{pmatrix} M.$$

For two inputs use $(axy + bx + cy + d)/(exy + fx + gy + h)$; its eight integer coefficients are closed under the same absorb-or-emit operations. Exact interval bounds certify a digit when every possible value has the same floor. Interval-valued output is the safe fallback at integer boundaries where a digit-only process can stall.

Calendar design. If a tropical year is $365 + \alpha$ days, a cycle with p leap days in q years asks for $p/q \approx \alpha$. For $\alpha \approx 0.2421988$, the convergents begin

$$\frac{1}{4}, \quad \frac{7}{29}, \quad \frac{8}{33}, \quad \frac{31}{128}.$$

The Julian calendar uses $1/4$. Requiring $q = 100Q$ means approximating 100α; its convergent $97/4$ yields the Gregorian rate $97/400$.

Planetary lapping time. If an inner planet takes V days for one Sun-centered lap and Earth takes $E > V$ days, the waiting time S between successive alignments (the *synodic period*) satisfies

$$\frac{1}{S} = \frac{1}{V} - \frac{1}{E}, \qquad \frac{S}{E} = T(E/V), \qquad T(r) = \frac{1}{r-1}.$$

For Earth and Venus, $E/V \approx 13/8$ and $S/E \approx 8/5$; the five successive conjunction directions advance by about 3/5 of a turn and draw a near-pentagram. The Dresden Codex uses the exact nominal calendar identities

$$5 \cdot 584 = 8 \cdot 365 = 2920, \qquad 65 \cdot 584 = 104 \cdot 365 = 146 \cdot 260 = 37960.$$

Pell bridge. If $p^2 - Dq^2 = \pm 1$, then

$$\left|\sqrt{D} - p/q\right| = \frac{1}{q^2(\sqrt{D} + p/q)} < \frac{1}{2q^2},$$

so Legendre makes p/q a convergent. Conversely, the period of $\sqrt{D}$ produces Pell solutions. A solution $u^2 - Dv^2 = 1$ gives the generator $\left(\begin{smallmatrix} u & Dv \\ v & u \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$.

Base-b expansion. Long division evolves by $br_n = qd_{n+1} + r_{n+1}$ with $0 \le r_{n+1} < q$. Repeated remainders make every rational expansion eventually periodic. A reduced fraction m/q is purely periodic exactly when $\gcd(q, b) = 1$; the reverse direction follows from Euler's theorem.

Linear congruence. For $ax \equiv b \pmod{n}$, first compute $d = \gcd(a, n)$. There are solutions iff $d \mid b$. Divide by d, reduce the modulus to n/d, and invert the new coefficient.

CRT. With $N = \prod n_i$, take $N_i = N/n_i$ and choose $M_i = N_i^{-1} \pmod{n_i}$. Then $x = \sum a_i N_i M_i \pmod{N}$.

Proof moves worth naming

- **Minimal counterexample / strong induction**: factor a composite object into smaller ones.
- **Bézout insertion**: replace 1 by $ax + by$ to transfer divisibility.
- **Exponent comparison**: use unique factorization or v_p to compare products and quotients.
- **Möbius monotonicity**: compute the derivative of $(at + b)/(ct + d)$ to determine how a continued-fraction tail moves its value.
- **Equal-block partition**: cosets and rotation orbits turn a count into divisibility.
- **Count two ways**: binomial identities become transparent when both sides count the same set.
- **Logarithmic weighting**: products of primes become the sums ϑ and ψ.
- **Dyadic decomposition**: split a range into $(x/2, x], (x/4, x/2], \ldots$ and sum a geometric series.

The two Fermat proofs

	Necklace proof	Group proof
Objects	a^p labeled colorings	nonzero residues modulo p
Structure	cyclic rotation orbits	group $(\mathbb{Z}/p\mathbb{Z})^\times$
Exceptional part	a constant colorings	identity/powers inside cyclic subgroups
Equal blocks	nonconstant orbits have size p	element orders divide $p - 1$
Conclusion	$a^p \equiv a \pmod{p}$	$a^{p-1} \equiv 1 \pmod{p}$ for $p \nmid a$

Further reading

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Course packet designed as an adaptable teaching draft. Examples, exercises, and diagrams are original to this packet.